Danish Technical University

10060 Physics summary

Comprehensive Study Notes from spring 2025 to fall 2025

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This document is a concise compilation of theoretical and applied physics concepts, formulas, and diagrams, covering key areas including mechanics, thermodynamics, electricity, magnetism, and modern physics.

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Contents

1 Measurement and Uncertainties

1.1 Why Uncertainty Matters

Every measurement has a limited precision. Understanding and reporting uncertainty lets us judge whether two results agree and how reliable our data are.

Types of Errors

- **Random errors:** fluctuate in sign and magnitude; reduced by repeated measurement.
- **Systematic errors:** constant bias due to calibration or method; not reduced by repetition.

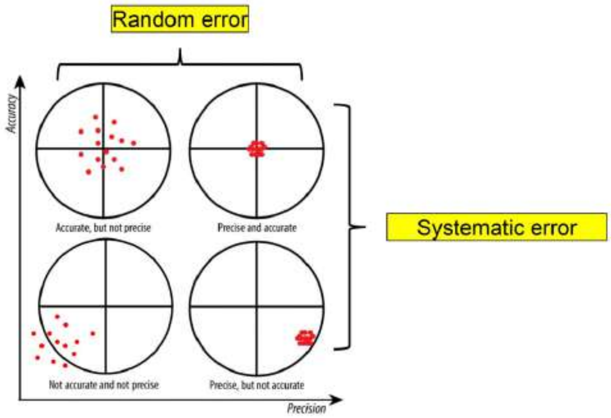


Figure 1: Examples of errors in measurements

1.2 Significant Figures and Rounding

- All non-zero digits are significant; zeros between or after decimals count.
- Round uncertainty to one significant digit (two if first is 1–3).
- Match digits of the value and its uncertainty: $L = 10.0 \pm 0.3$ m.

1.3 Error Propagation

For a quantity $f(x, y, \dots)$ with uncertainties δx, δy:

$$\delta f = \sqrt{\left(\frac{\partial f}{\partial x}\delta x\right)^2 + \left(\frac{\partial f}{\partial y}\delta y\right)^2 + \dots}$$

Addition/Subtraction

$$\delta q = \sqrt{(\delta x)^2 + (\delta y)^2}$$

Multiplication/Division

$$\frac{\delta q}{q} = \sqrt{\left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta y}{y}\right)^2}$$

Dependent errors arise from the same source and do not combine in quadrature.

Example of finding uncertainties and components of uncertainties in Python,

```
from uncertainties import *

h0 = ufloat(1.60, 0.05)
v0 = ufloat(4.20, 0.05)
g = ufloat(9.82, 0.01)
t = (-v0 - (v0**2 + 2*g*h0)**(0.5)) / -g

t, t.error_components()
```

1.4 Repeated Measurements

For N measurements x_i:

$$\bar{x} = \frac{1}{N}\sum_i x_i, \quad \sigma = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{N-1}}, \quad \delta\bar{x} = \frac{\sigma}{\sqrt{N}}$$

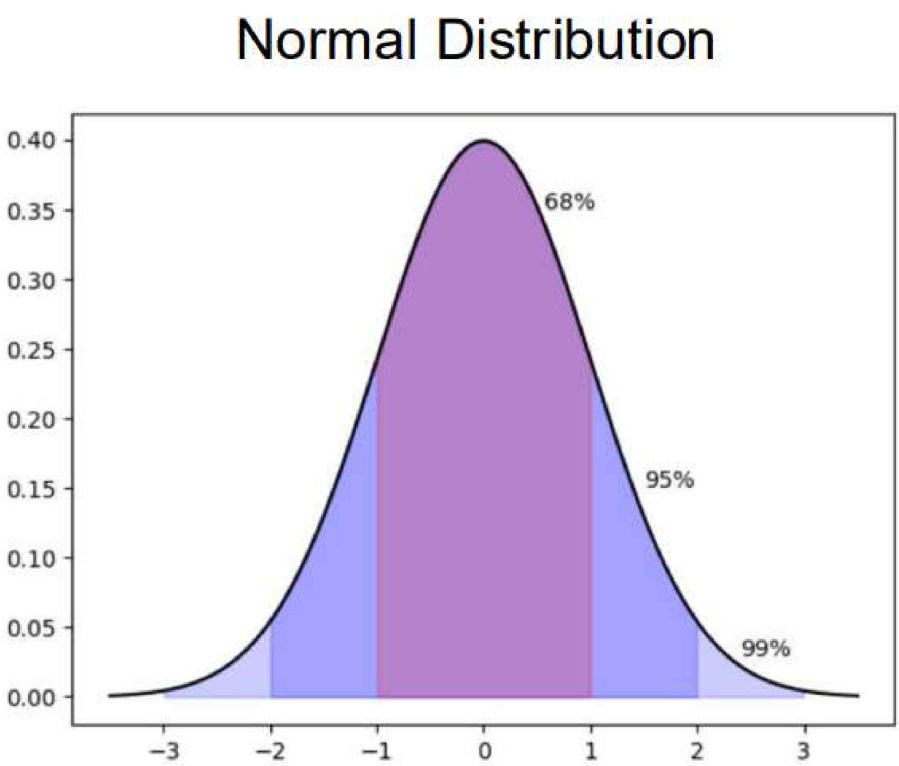
σ is the standard deviation; $\delta\bar{x}$ is the **standard deviation of the mean (SDOM)**.

1.5 Probability Distributions and Confidence

Measurements often follow a normal (Gaussian) distribution:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

68% of values lie within $\pm 1\sigma$, 95% within $\pm 2\sigma$, 99% within $\pm 3\sigma$.



Normal distribution and confidence intervals (68–95–99 rule).

1.6 Accuracy, Precision, and Common Errors

- **Accuracy:** closeness to the true value.

- **Precision:** repeatability of results.
- **Parallax error:** reading analog scale from an angle.
- **Digital rounding error:** limited by instrument resolution.

2 Experimental Method and Dimensional Analysis

2.1 Types of Experiments

- **Observational:** explore phenomena without predictions.
- **Test:** test hypotheses and compare with theoretical predictions.
- **Application:** determine unknown quantities using established laws.

2.2 Designing an Experiment

Scientific Method in Six Steps

1. Define the research question.
2. Formulate a hypothesis.
3. Plan setup — variables, instruments, methods.
4. Acquire data.
5. Analyze and interpret results.
6. Report and evaluate.

2.3 Comparing Measurements

Standard Score Test

$$z = \frac{x - x_{\text{ref}}}{\delta x}, \qquad \begin{cases} |z| \leq 2 & \text{consistent,} \\ 2 < |z| \leq 3 & \text{gray region,} \\ |z| > 3 & \text{inconsistent.} \end{cases}$$

For two independent measurements $x \pm \delta x$ and $y \pm \delta y$:

$$z = \frac{x - y}{\sqrt{\delta x^2 + \delta y^2}}$$

2.4 Dimensional Analysis and Natural Scales

Each physical quantity can be expressed as a combination of base dimensions (M, L, T,...). Dimensional analysis identifies characteristic scales that simplify equations.

Dimensional Homogeneity

$$[F] = MLT^{-2}$$

Natural Scales (Pendulum Example)

$$\tau = \sqrt{\frac{l}{g}}$$

Dimensionless form of a relation simplifies comparison between systems:

$$\frac{T}{\sqrt{l/g}} = f\left(\frac{A}{l}\right)$$

2.5 Example — Measuring the Acceleration of Gravity

Assuming motion with constant acceleration:

$$h = \frac{1}{2}gt^2$$

Plotting h versus t^2 yields a straight line; slope $= \frac{1}{2}g$.

The constant k obtained experimentally ($g = k\,h/t^2$) should be close to 2.

2.6 Air Resistance (Advanced)

When drag is proportional to velocity ($F = kv$), dimensional analysis gives:

$$\frac{tk}{m} = f\left(\frac{m^2ghk^2}{}\right)$$

leading to dimensionless groups used to collapse data from various masses and heights.

3 Units of Measurement

3.1 SI Base Quantities and Units

Quantity	Symbol	Unit (SI)	Unit Symbol
Length	l	meter	m
Mass	m	kilogram	kg
Time	t	second	s
Electric current	I	ampere	A
Thermodynamic temperature	T	kelvin	K
Amount of substance	n	mole	mol
Luminous intensity	I_v	candela	cd

3.2 Notes

- Use roman (upright) fonts for unit symbols (m, s), italic for variables (t, v, a).
- Products of units use a middle dot or a thin space: N m, kg m/s^2.
- Powers and ratios: m^2, m/s, m/s^2.

3.3 Common SI Prefixes

Prefix	Symbol	Factor
giga	G	10^9
mega	M	10^6
kilo	k	10^3
hecto	h	10^2
deca	da	10^1
deci	d	10^{-1}
centi	c	10^{-2}

Prefix	Symbol	Factor
tera	T	10^{12}
peta	P	10^{15}
exa	E	10^{18}
zetta	Z	10^{21}
yotta	Y	10^{24}
pico	p	10^{-12}
femto	f	10^{-15}
atto	a	10^{-18}
zepto	z	10^{-21}
yocto	y	10^{-24}

3.4 Unit Conventions (Quick Reference)

- **Force:** $\mathrm{N} = \mathrm{kg\,m/s^2} \quad (\mathbf{F} = m\mathbf{a})$
- **Work/Energy:** $\mathrm{J} = \mathrm{N\,m} = \mathrm{kg\,m^2/s^2}$
- **Power:** $\mathrm{W} = \mathrm{J/s}$
- **Charge:** $\mathrm{C} = \mathrm{A\,s}$
- **Potential:** $\mathrm{V} = \mathrm{W/A} = \mathrm{J/C}$

4 Kinematics

Kinematics describes how objects move, without reference to the forces that cause the motion. Key quantities are position $x(t)$, velocity $v(t)$, and acceleration $a(t)$.

4.1 Approach to Solving Physics Problems

Problem-Solving Strategy

1. **Understand the problem:** Identify knowns and unknowns.
2. **Sketch and simplify:** Draw a diagram, choose coordinates, neglect irrelevant details.
3. **Apply physics:** Select the relevant kinematic relations.
4. **Solve algebraically, then numerically.**
5. **Check reasonableness:** Units, signs, and limits.
6. **Generalize:** Identify how results change for different conditions.

4.2 Types of Motion in One Dimension

Constant Position

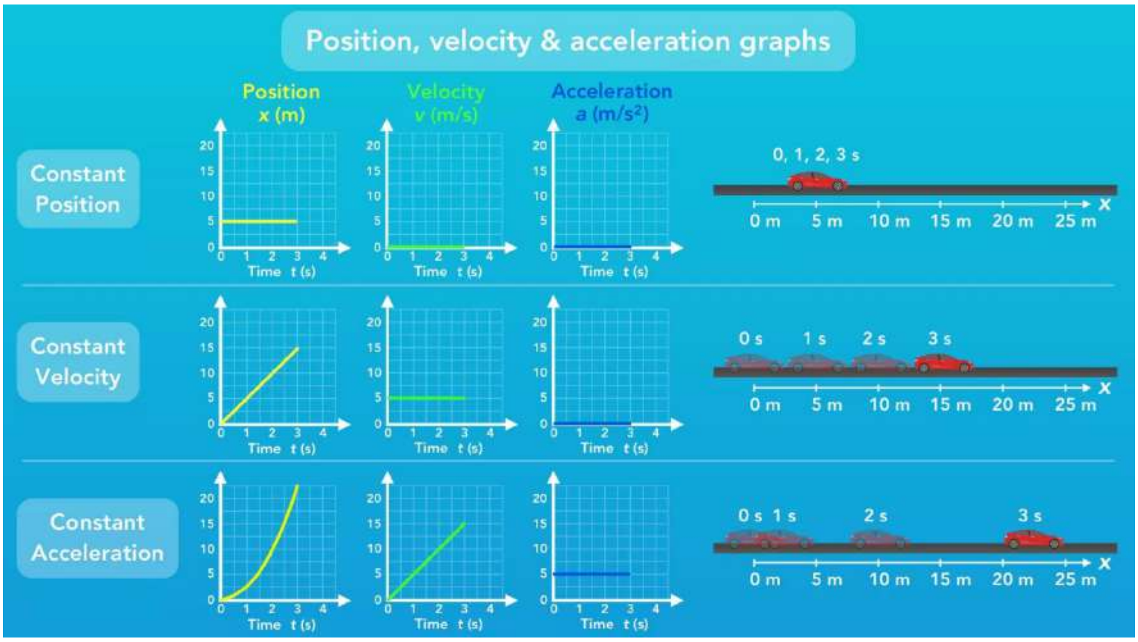
$$x(t) = x_0$$

Constant Velocity

$$x(t) = x_0 + v_0 t$$

Constant Acceleration

$$x(t) = x_0 + v_0 t + \tfrac{1}{2}at^2$$
$$v(t) = v_0 + at$$
$$v^2 = v_0^2 + 2a(x - x_0)$$



Typical position, velocity, and acceleration vs. time graphs for uniform acceleration.

4.3 Example: Free Fall

Objects in free fall near Earth's surface experience constant downward acceleration $g = 9.82$ m/s^2, independent of their mass (neglecting air resistance):

$$y(t) = y_0 + v_0 t - \tfrac{1}{2}gt^2, \qquad v(t) = v_0 - gt$$

Practical application: The depth of a well can be estimated by dropping a rock and timing the fall:

$$h = \tfrac{1}{2}gt^2$$

Example: if the sound of impact is heard after 2.0 s, $h \approx 19.6$ m (neglecting sound delay).

4.4 Types of Motion in Two Dimensions

Motion in two dimensions can be resolved into independent horizontal and vertical components:

$$\begin{cases} x(t) = x_0 + v_{0x}t, \\ y(t) = y_0 + v_{0y}t - \tfrac{1}{2}gt^2, \end{cases} \qquad \begin{cases} v_x = v_{0x}, \\ v_y = v_{0y} - gt. \end{cases}$$

4.5 Projectile Motion

Projectile motion is an example of 2D motion under constant acceleration (gravity).

Time of Flight

$$T = \frac{2v_0 \sin\theta_0}{g}$$

Range

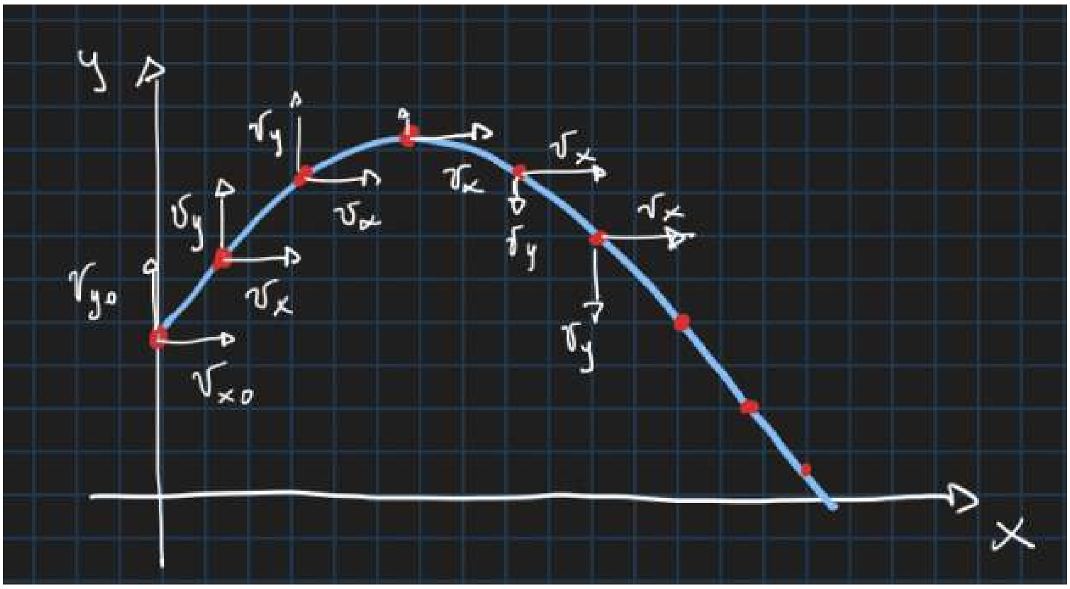
$$R = \frac{v_0^2 \sin(2\theta_0)}{g}$$

Maximum Height

$$y_{\max} = \frac{v_0^2 \sin^2\theta_0}{2g}$$

Trajectory

$$y(x) = x\tan\theta_0 - \frac{gx^2}{2v_0^2\cos^2\theta_0}$$



Projectile trajectory with velocity components.

4.6 Uniform Circular Motion

An object moving in a circle of radius r at constant speed v has angular speed $\omega = v/r$ and experiences a centripetal acceleration directed toward the center.

Angular Position

$$\theta(t) = \omega t$$

Centripetal Acceleration

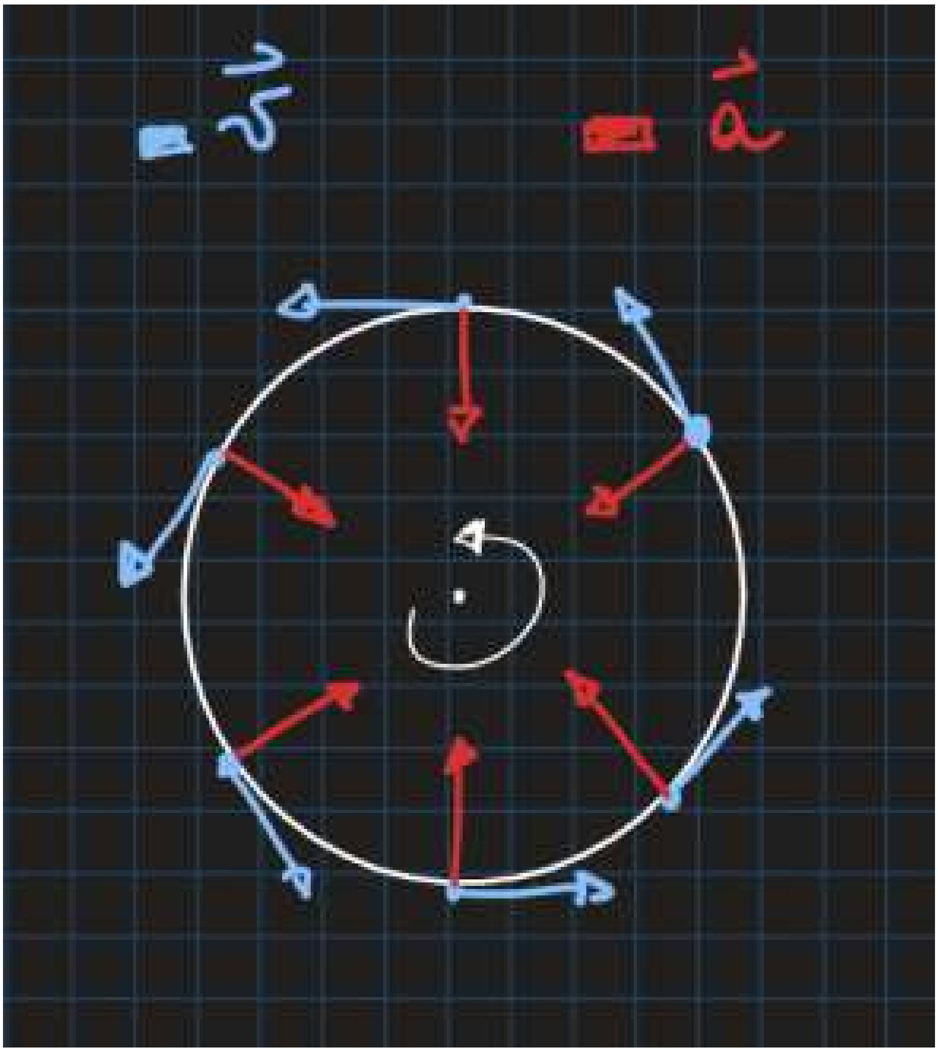
$$a_c = \frac{v^2}{r} = r\omega^2$$

Linear Speed

$$v = r\omega$$

Period

$$T = \frac{2\pi}{\omega} = \frac{2\pi r}{v}$$



Velocity and acceleration in uniform circular motion.

Example for a pendulum bob on a massless, inextensible string, a the top of the circle, the tension must be ≥ 0

4.7 Relative Motion

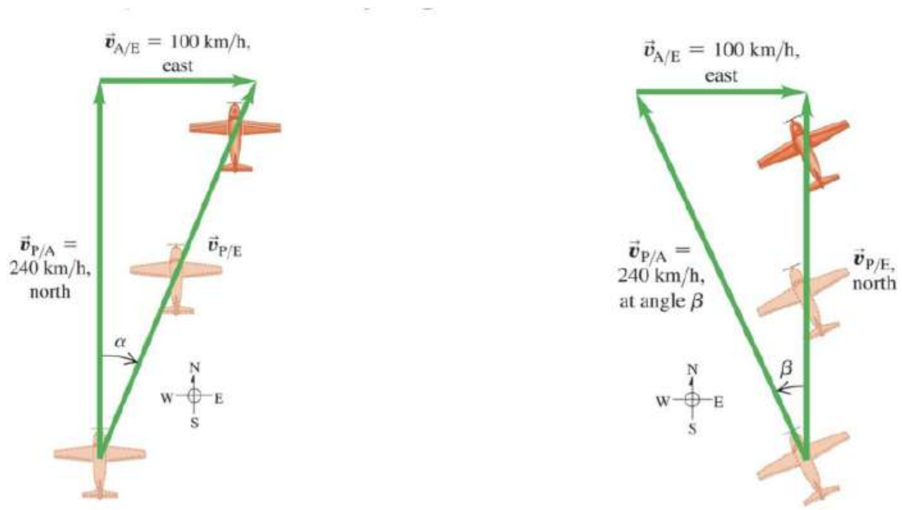
Relative motion describes how the motion of one object appears to an observer moving with another object.

Velocity Addition (Vector Form)

$$\mathbf{v}_{A/C} = \mathbf{v}_{A/B} + \mathbf{v}_{B/C}$$

Example: Flying in a Crosswind
An airplane has airspeed $v_{A/B} = 200$ km/h east and faces a wind of $v_{B/C} = 50$ km/h north. The ground velocity is

$$\mathbf{v}_{A/C} = 200\,\hat{\mathbf{i}} + 50\,\hat{\mathbf{j}}, \quad |\mathbf{v}_{A/C}| = 206 \text{ km/h}, \quad \theta = 14^\circ \text{ N of E.}$$



Vector addition of airplane and wind velocities.

4.8 Numerical Equations of Motion

When analytic solutions are impossible, motion can be simulated numerically.

Euler's Method

$$x(t + \Delta T) \approx x(t) + v(t)\Delta t$$
$$v(t + \Delta T) \approx v(t) + a(t)\Delta t$$

These update position and velocity step-by-step with time step ΔT.

Euler–Cromer Methods

$$v(t + \Delta T) \approx v(t) + a(t)\Delta t$$
$$x(t + \Delta T) \approx x(t) + v(t + \Delta t)\Delta t$$

These update position and velocity step-by-step with time step ΔT.

In Python: numerical solvers like `scipy.integrate.solve_ivp(f, t_span, y0)` automate higher-order schemes and handle event detection (e.g., projectile hitting ground).

```
import numpy as np
from scipy.integrate import solve_ivp
t1 = 0.0
t2 = 10000.0
N = 1000000
t = np.linspace(0,10000.0,N +1)
R = 6370000
g = 9.81
def f(t,y):
    x,v = y
    dxdt = v
    dvdt = -g*x/R
    return [dxdt,dvdt]
def maxima(t,y):
```

```
    x,v = y
    return x
maxima.direction = -1
sol = solve_ivp(f,[t[0],t[-1]],[6370000,0.0],t_eval=t,events=[maxima],rtol=1e-8,atol=1e-8)
print(sol)
t = sol.t
x = sol.y[0]
v = sol.y[1]
```

```
import numpy as np
from scipy.integrate import solve_ivp

# constants
g = 9.82
l = 1.0

# equation of motion
def rhs(t, y):
    theta, omega = y
    dtheta_dt = omega
    domega_dt = -(3*g/(2*l)) * np.sin(theta)
    return [dtheta_dt, domega_dt]

# initial conditions: released from rest at 45 degrees
theta0 = np.deg2rad(45)
omega0 = 0.0
y0 = [theta0, omega0]

# integrate long enough for one full period
sol = solve_ivp(
    rhs,
    [0, 10],
    y0,
    max_step=1e-3,
    rtol=1e-9,
    atol=1e-12
)

t = sol.t
theta = sol.y[0]

# find first downward crossing of theta = 0
crossings = np.where((theta[:-1] > 0) & (theta[1:] <= 0))[0]
i = crossings[0]

# linear interpolation for accurate crossing time
t0, t1 = t[i], t[i+1]
th0, th1 = theta[i], theta[i+1]
t_cross = t0 - th0 * (t1 - t0) / (th1 - th0)

# full period
T = 4 * t_cross
print("Oscillation period T =", T)
```

4.9 Oscillations and Resonance

Damped and Forced Oscillator

$$m\ddot{x} + b\dot{x} + kx = A\sin(\omega t)$$

- $b = 0, A = 0$: simple harmonic motion, $\omega_0 = \sqrt{k/m}$.
- $b > 0, A = 0$: damped oscillation, amplitude $\propto e^{-bt/2m}$.
- $A \neq 0$: forced oscillation; resonance when $\omega \approx \omega_0$.

4.10 Key Notes

- Kinematic equations apply separately to each coordinate direction.
- At the top of a projectile's path: $v_y = 0$.
- For symmetric trajectories: $t_{\text{up}} = t_{\text{down}} = \frac{1}{2}T$.
- In circular motion, velocity is tangent to the path; acceleration points radially inward.
- Use vector diagrams to visualize 2D and 3D motion clearly.

5 Forces

Forces are vector quantities measured in newtons (N). They describe interactions that can change an object's motion or shape.

Kinematics describes how objects move; **dynamics** explains *why* they move, relating motion to the forces that cause it.

Inertial and Non-Inertial Frames

Newton's laws are valid only in **inertial frames** (no acceleration). Accelerating frames introduce *fictitious forces* (e.g. centrifugal, Coriolis) to account for apparent motion.

5.1 Drawing a Free-Body Diagram

1. Represent the object as a point at the origin of a coordinate system.
2. Include all external forces acting on the object (not the net force).
3. Resolve all forces into x and y components.
4. Draw a separate diagram for each interacting object.

5.2 Newton's Laws of Motion

First Law (Law of Inertia)

A body remains at rest or moves with constant velocity unless acted on by a net external force.

Second Law (Force–Acceleration Relation)

$$\sum \mathbf{F} = m\mathbf{a}$$

Acceleration is directly proportional to the net external force and inversely proportional to mass.

Internal vs External Forces: Only *external* forces contribute to $\sum \mathbf{F} = m\mathbf{a}$. Internal forces within a system (e.g. molecular bonds) cancel pairwise.

Third Law (Action–Reaction)

When body A exerts a force on body B, B exerts an equal and opposite force on A:

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA}$$

Component form of Newton's Second Law:

$$\sum F_x = ma_x, \qquad \sum F_y = ma_y, \qquad \sum F_z = ma_z$$

5.3 Fundamental Forces of Nature

Force	Example	Relative Strength / Range
Gravitational	Solar systems, galaxies	Weakest; infinite range
Weak Nuclear	Radioactive decay	Short range ($\sim 10^{-18}$ m)
Electromagnetic	Atoms, molecules	Stronger; infinite range
Strong Nuclear	Atomic nuclei	Strongest; $\sim 10^{-15}$ m range

5.4 Contact and Restoring Forces

- **Normal Force (N)** — acts perpendicular to a surface.
- **Frictional Force (f)** — acts parallel to a surface, opposing motion.
- **Tension (T)** — pulling force transmitted by strings, ropes, etc.
- **Spring Force (Hooke's Law)** — restoring force proportional to displacement:

$$\mathbf{F}_{\text{spring}} = -k\,\Delta x\,\hat{\mathbf{x}}$$

where k is the spring constant (N/m) and Δx is the stretch/compression.

5.5 Friction

The force of friction is computed as:

$$f = \mu N$$

where μ is the coefficient of friction and N is the normal force.

- **Static Friction** (object not moving):

$$f_s \leq \mu_s N, \qquad f_{s,\max} = \mu_s N$$

- **Kinetic Friction** (object sliding):

$$f_k = \mu_k N$$

Typical values: $\mu_s > \mu_k$, and both depend on surface materials.

Microscopic origin: friction arises from interlocking surface asperities and microscopic adhesive forces between contact points.

Static vs kinetic friction (exam note) When both coefficients μ_s and μ_k are given:

- Static friction adjusts to prevent motion, up to a maximum

$$f_s^{\max} = \mu_s N.$$

- First assume the object is at rest and compute the friction force required for equilibrium.
- If the required friction satisfies $f \leq \mu_s N$, the object does not move and static friction applies.
- If $f > \mu_s N$, static friction cannot hold; the object moves and kinetic friction $f_k = \mu_k N$ must be used.

5.6 Weight

Weight is the gravitational force acting on an object:

$$\mathbf{W} = m\mathbf{g}$$

where $\mathbf{g}$ is the acceleration due to gravity ($9.81\,\mathrm{m/s^2}$ near Earth's surface).

5.7 Drag Forces

For objects moving through a fluid:

Quadratic Drag (High Speed)

$$F_D = \frac{1}{2}C\rho A v^2$$

C — drag coefficient, ρ — fluid density, A — cross-sectional area.

Stokes' Law (Low Speed

$$F_s = 6\pi r \eta v$$

r — radius of sphere, η — viscosity of fluid.

Terminal velocity: when $F_D = W$, acceleration ceases:

$$v_t = \sqrt{\frac{2mg}{C\rho A}}$$

5.8 Centripetal Force

For circular motion of radius r:

$$F_c = ma_c = \frac{mv^2}{r}$$

The centripetal force acts toward the center of the circle.

5.9 Conservative vs Non-Conservative Forces

- **Conservative:** work is path-independent, energy is recoverable (e.g., gravity, spring force).
- **Non-Conservative:** work depends on the path and energy is dissipated (e.g., friction, drag).

Centripetal Force on a Banked Curve

For a car of mass m rounding a curve of radius r banked at angle θ:

$$v = \sqrt{rg\tan\theta}.$$

No friction is required at this speed; higher or lower speeds need frictional force.

5.10 Simple Harmonic Motion

In simple harmonic motion, the acceleration of the system, and therefore the net force, is proportional to the displacement and acts in the opposite direction of the displacement.

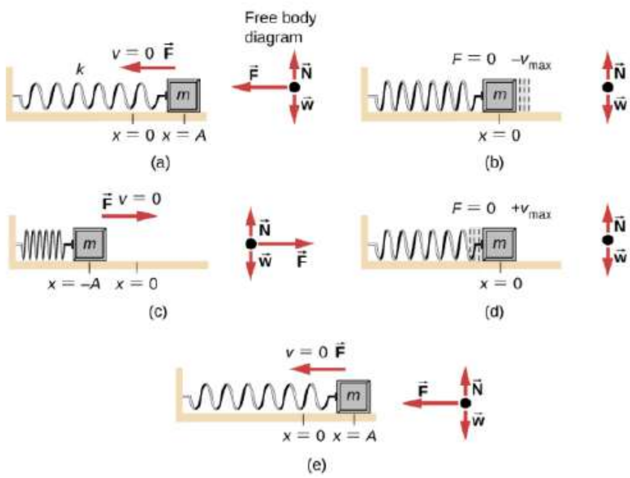


Figure 2: Simple Harmonic Motion in a spring

The position vs time equation for a simple harmonic motion is given as,

$$x(t) = A\cos\left(\frac{2\pi}{T}t\right) = A\cos(\omega t)$$

Where $\omega = \frac{d\theta}{dt} = 2\pi/T$ is the angular frequency

The velocity vs time equations is given as,

$$v(t) = \frac{dx}{dt} = \frac{d}{dt}(A\cos(\omega t + \phi)) = -A\omega\sin(\omega t + \phi) = -v_{\max}\sin(\omega t + \phi)$$

Because the sine function oscillates between -1 and 1, the maximum velocity is the amplitude times the angular frequency $v_{\max} = A\omega$

The acceleration of the mass on the spring can be found by taking the time derivative of the velocity as,

$$a(t) = \frac{dv}{dt} = \frac{d}{dt}(-A\omega\sin(\omega t + \phi)) = -A\omega^2\cos(\omega t + \phi) = -a_{\max}\cos(\omega t + \phi)$$

The maximum acceleration is $a_{\max} = A\omega^2$

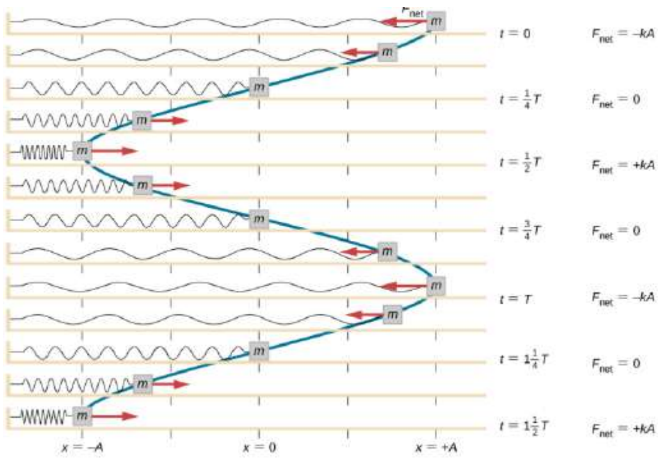


Figure 3: Simple Harmonic Motion, Spring, time, force

6 Energy, Work, and Power

Energy is the ability to perform work. It can exist in several forms: mechanical (kinetic and potential), thermal, chemical, and others. The SI unit of energy and work is the **joule (J)**:

$$1 \text{ J} = 1 \text{ N m} = 1 \text{ kg m}^2/\text{s}^2$$

6.1 Kinetic Energy

Definition

$$K = \frac{1}{2}mv^2$$

Kinetic energy is the energy associated with motion. It is proportional to the mass of the object and to the square of its speed.

6.2 Potential Energy

Potential energy is energy due to position or configuration.

Gravitational Potential Energy

$$U_g = mgh$$

Elastic Potential Energy

$$U_{el} = \frac{1}{2}kx^2$$

6.3 Mechanical Energy

Mechanical energy is the sum of kinetic and potential energies:

$$E_{\text{mech}} = K + U$$

If only conservative forces act, mechanical energy is conserved:

$$E_{\text{mech, i}} = E_{\text{mech, f}}$$

6.4 Thermal Energy

Thermal energy represents the microscopic kinetic energy of molecules. It manifests as a change in temperature:

$$W_{friction\ force} = Q = mc\Delta T$$

where m is mass, c is specific heat capacity, and ΔT is the temperature change.

6.5 Work

Work is the transfer of energy to or from an object by means of a force and a displacement.

Uniform Force

$$W = \mathbf{F} \cdot \Delta\mathbf{r} = F\Delta r \cos\theta$$

Non-Uniform Force (Variable with Position)

$$W = \int_{x_1}^{x_2} F(x)\, dx$$

The work is the area under the force–position graph.

Spring Work

$$W_{\text{spring}} = \frac{1}{2}kX^2$$

where X is the displacement from equilibrium.

6.6 Work–Energy Theorem

$$W_{\text{net}} = \Delta K$$

The net work done on a particle equals the change in its kinetic energy. For conservative forces:

$$W_{\text{cons}} = -\Delta U$$

and thus:

$$\Delta E_{\text{mech}} = 0 \qquad \text{if only conservative forces act.}$$

Sign of Work:

- $W > 0$ — force adds energy to the object (speeding up).
- $W < 0$ — force removes energy (friction, drag).
- $W = 0$ — force perpendicular to motion (e.g. uniform circular motion).

Example – Power Output in a Pull-Up: A student of mass 70 kg lifts themselves 0.4 m in 1 s:

$$P = \frac{mgh}{t} = \frac{70 \times 9.81 \times 0.4}{1} \approx 275 \text{ W}.$$

Work for a Non-Uniform (Spring) Force:

$$W = \int_0^x F(x')\, dx' = \int_0^x kx'\, dx' = \tfrac{1}{2}kx^2.$$

6.7 Power

Power is the rate at which work is done or energy is transferred.

Average Power

$$P_{\text{avg}} = \frac{W}{\Delta t}$$

Instantaneous Power

$$P = \frac{\mathrm{d}W}{\mathrm{d}t} = \mathbf{F} \cdot \mathbf{v}$$

The SI unit of power is the **watt (W)**:

$$1\ \text{W} = 1\ \text{J/s}$$

6.8 Potential Energy and Conservative Forces

For conservative forces, potential energy is defined by:

$$\Delta U = -W_{\text{cons}} = -\int_{x_1}^{x_2} F(x)\,dx$$

and conversely:

$$F(x) = -\frac{\mathrm{d}U}{\mathrm{d}x}$$

This relationship connects the potential energy function to the force acting on an object.

6.9 Summary of Energy Relationships

$$\begin{aligned} W_{\text{net}} &= \Delta K \\ W_{\text{cons}} &= -\Delta U \\ E_{\text{mech}} &= K + U = \text{constant (if only conservative forces)} \\ P &= \frac{\mathrm{d}W}{\mathrm{d}t} = \mathbf{F} \cdot \mathbf{v} \end{aligned}$$

7 Momentum, Impulse, and Collisions

7.1 Linear Momentum and Impulse

Linear Momentum

$$\mathbf{p} = m\mathbf{v}$$

Impulse

$$\mathbf{J} = \int_{t_1}^{t_2} \mathbf{F}\,dt = \Delta \mathbf{p}$$

Units: N s. The impulse on an object equals its change in momentum.

7.2 Conservation of Momentum

For an isolated system (no net external force),

$$\sum \mathbf{p}_{\text{i}} = \sum \mathbf{p}_{\text{f}}.$$

Types of collisions:

- **Elastic:** momentum and kinetic energy conserved.
- **Inelastic:** momentum conserved, kinetic energy not conserved.
- **Perfectly inelastic:** objects stick together.

7.3 Center of Mass (CM)

$$\mathbf{r}_{\mathrm{CM}} = \frac{\sum_i m_i \mathbf{r}_i}{\sum_i m_i}, \qquad \mathbf{v}_{\mathrm{CM}} = \frac{\sum_i m_i \mathbf{v}_i}{\sum_i m_i}, \qquad \mathbf{a}_{\mathrm{CM}} = \frac{\sum_i m_i \mathbf{a}_i}{\sum_i m_i}, \qquad \mathbf{p}_{\mathrm{tot}} = M\,\mathbf{v}_{\mathrm{CM}}.$$

Only **external** forces can change $\mathbf{p}_{\mathrm{tot}}$ or $\mathbf{v}_{\mathrm{CM}}$.

7.4 1D Collision Quick Results

Two masses m_1, m_2 with initial velocities u_1, u_2 (elastic):

$$v_1 = \frac{(m_1 - m_2)u_1 + 2m_2u_2}{m_1 + m_2}, \quad v_2 = \frac{(m_2 - m_1)u_2 + 2m_1u_1}{m_1 + m_2}.$$

Perfectly inelastic (stick together): $v = \dfrac{m_1u_1 + m_2u_2}{m_1 + m_2}$.

7.5 Rocket Motion (Variable Mass)

For exhaust speed v_e (relative to the rocket), negligible external force:

$$v = v_e \ln\frac{m_0}{m}$$

(*Tsiolkovsky rocket equation*). With external force F_{ext},

$$F_{\mathrm{ext}} = m\frac{\mathrm{d}v}{\mathrm{d}t} + v_e\frac{\mathrm{d}m}{\mathrm{d}t}.$$

8 Rotational Motion

8.1 Angular Quantities

Angular Displacement

$$\theta = \frac{s}{r}$$

Angular Acceleration

$$\alpha = \frac{\mathrm{d}\omega}{\mathrm{d}t}$$

Angular Velocity

$$\omega = \frac{\mathrm{d}\theta}{\mathrm{d}t}$$

Linear–rotational links: $v = r\omega, \quad a_t = r\alpha, \quad a_c = r\omega^2$.

Rotational Kinematics (Constant α)

$$\omega = \omega_0 + \alpha t, \qquad \theta = \theta_0 + \omega_0 t + \tfrac{1}{2}\alpha t^2, \qquad \omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0).$$

8.2 Moment of Inertia

The moment of inertia of a body for a given rotation axis is given by,

$$I = \sum_i m_i r_i^2 = \int r^2 \, dm$$

Where r is the distance to the axis of rotation.

This is true for every type of body.

The mass of a surface is equal to the surface density times area.

The mass of a volume is equal to the volume density times the volume.

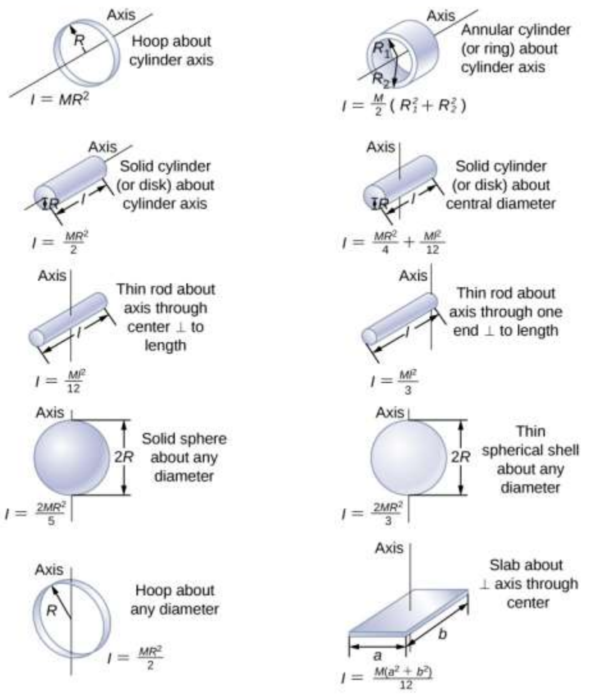


Figure 4: Moment of inertia for various bodies (1)

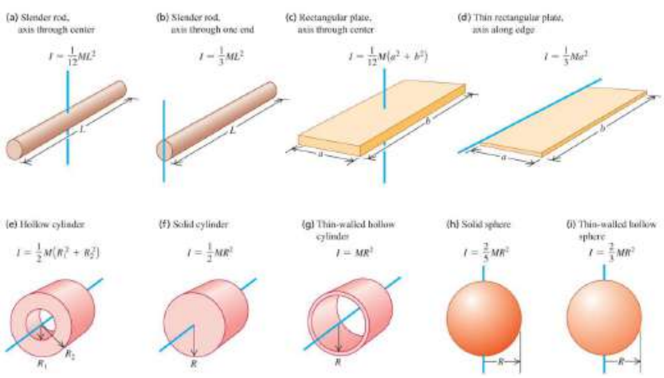


Figure 5: Moment of inertia for various bodies (2)

Parallel-Axis (Steiner) Theorem

Let m be the mass of an object, and let d be the distance from an axis through the object's center of mass to a new axis. Then we have,

$$I_{\text{parallel-axis}} = I_{\text{CM}} + Md^2$$

8.3 Rolling Without Slipping

$$v_{\mathrm{CM}} = \omega R, \qquad a_{\mathrm{CM}} = \alpha R$$

Total energy for a rolling body (with height h):

$$E = \tfrac{1}{2} I\omega^2 + \tfrac{1}{2} M v_{\mathrm{CM}}^2 + Mgh.$$

8.4 Rotational Kinetic Energy

For an arbitrary point on the circumference of a wheel, the rotational kinetic energy is,

$$K_{\mathrm{rot}} = \sum_i \frac{1}{2} m_i r_i^2 \omega^2$$

The rotational kinetic energy of a rigid body rotating around an axis can therefore be written as,

$$K = \frac{1}{2} I\omega^2$$

The expression $K = \frac{1}{2}mv^2$ is valid only for pure translation. For a rigid body rotating about a fixed axis, $K = \frac{1}{2} I\omega^2$.

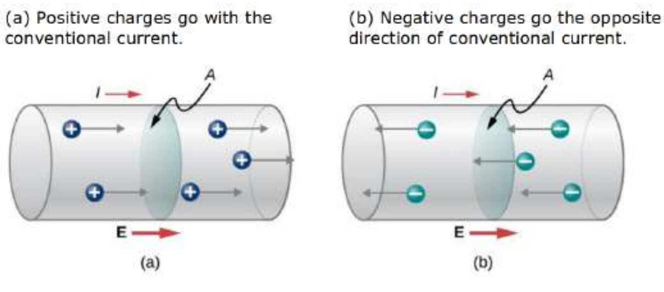
9 Electric Current and Circuits

9.1 Electric Current

Current is the rate of flow of charge through a surface:

$$I = \lim_{\Delta t \to 0} \frac{\Delta Q}{\Delta t} = \frac{dq}{dt}, \qquad 1\ \mathrm{A} = 1\ \mathrm{C/s}.$$

Direction of conventional current is the motion of **positive** charges and is opposite to electron flow.



9.2 Model of Conduction in Metals

Conduction in metals occurs through the motion of free electrons in response to an applied electric field. While individual electrons drift slowly, the electric signal propagates at nearly the speed of light through the conductor.

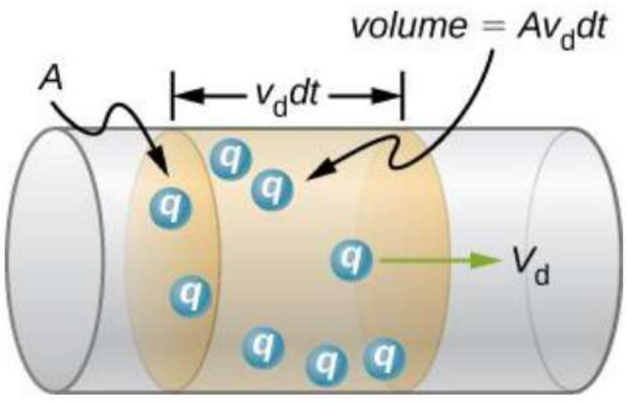


Figure 6: Electron drift in a conductor: charge carriers with density n and drift velocity v_d produce a current $I = qnAv_d$.

$$v_{\text{signal}} \sim 10^8 \text{ m/s}$$

Electrons experience many collisions with atoms, producing a net *drift velocity* in the direction opposite to the electric field.

$$I = \frac{dQ}{dt} = qnAv_d$$
$$v_d = \frac{I}{nqA}$$

where:

- I — current (A)
- n — number density of charge carriers (electrons)
- q — charge of each carrier (C)
- A — cross-sectional area of the conductor (m^2)
- v_d — average drift velocity (m/s)

In typical metallic circuits:

$$v_d \sim 10^{-4} \text{ m/s}$$

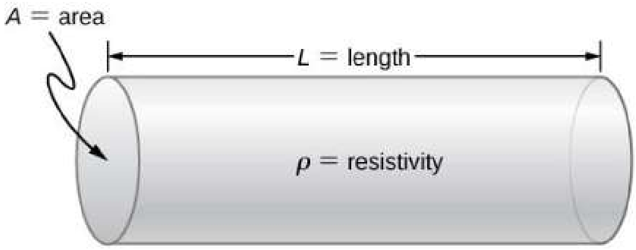
Although v_d is small, the **electric field signal** and energy transfer move through the circuit almost instantaneously.

9.3 Current Density

The current density $\vec{\mathbf{J}}$ is defined as the current passing through an infinitesimal cross-sectional area divided by that area.

$$\mathbf{J} = \frac{I}{A}\hat{\mathbf{n}}, \qquad I = \int \mathbf{J} \cdot d\mathbf{A}.$$

9.4 Resistivity and Resistance



For a uniform conductor of length L and cross-section A like the one above:

$$R = \rho \frac{L}{A}$$

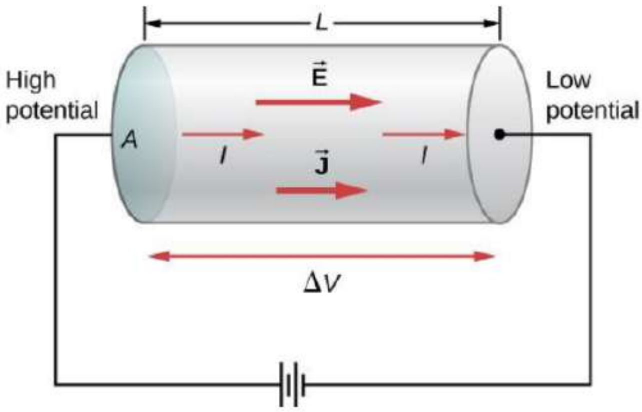
where ρ is the resistivity ($\Omega\,\mathrm{m}$).

Temperature dependence:

$$\rho = \rho_0[1 + \alpha(T - T_0)].$$

9.5 Ohm's Law

$$V = IR, \qquad \mathbf{J} = \sigma \mathbf{E}, \quad \sigma = \frac{1}{\rho}.$$



9.6 Electromotive Force (emf) and Batteries

A source of **electromotive force (emf)** maintains a potential difference by doing work on charges, moving them from lower to higher potential. It is defined as the work per unit charge:

Electromotive Force

$$\varepsilon = \frac{dW}{dq}$$

An ideal emf source keeps a fixed potential difference ε between its terminals, independent of the current.

Real Batteries and Internal Resistance

Real batteries have an internal resistance r. They are modeled as an ideal emf ε in series with r, connected to an external load resistance R:

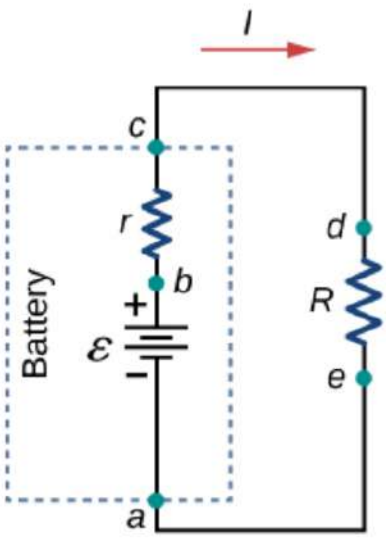


Figure 7: Circuit model of a real battery with emf ε, internal resistance r, and load resistance R.

If a current I flows in the circuit, there is a voltage drop $\Delta V = Ir$ *inside* the battery on the internal resistance and a voltage drop $\Delta V = IR$ on the external resistor. The terminal voltage (the voltage available to the external circuit) is therefore

Terminal Voltage of a Real Battery

$$V_{\text{terminal}} = \varepsilon - Ir$$

This behaviour is illustrated in the potential–position diagram below:

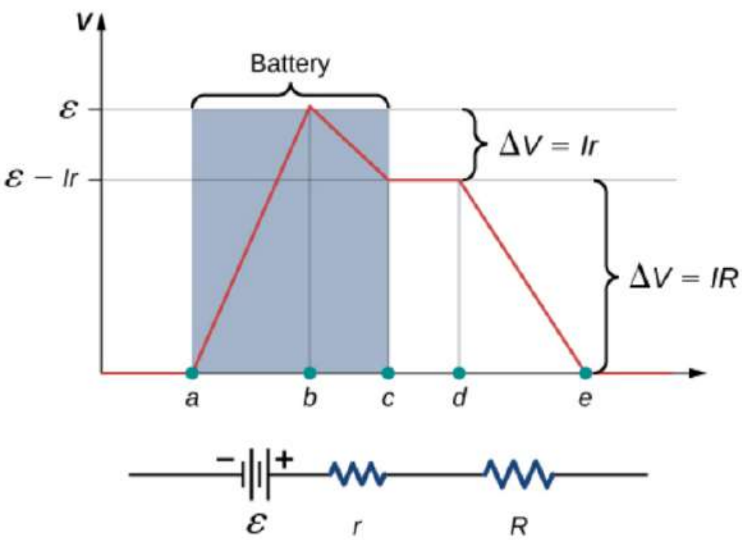


Figure 8: Voltage as a function of position around the circuit. The drop Ir occurs inside the battery (internal resistance), and the drop IR occurs across the external resistor.

For a simple circuit with total resistance $R + r$:

$$I = \frac{\varepsilon}{R + r}$$

- **Open circuit:** $I = 0 \Rightarrow V_{\text{terminal}} = \varepsilon$.

- **Loaded battery:** $I > 0 \Rightarrow V_{\text{terminal}} < \varepsilon$ due to the internal drop Ir.

Chemically, batteries separate charge so that one terminal is kept at higher potential than the other; the emf describes this energy supply to the circuit, independent of the detailed chemistry.

9.7 Electric Power

$$P = \frac{\Delta U}{\Delta t} = -\frac{\Delta QV}{\Delta t} = IV = I^2R = \frac{V^2}{R}.$$

9.8 Cost of electricity

$$E = \int Pdt \xrightarrow[\text{At constant power}]{} E = Pt \tag{1}$$

9.9 Energy Units

Electric energy consumption is often given in kilowatt-hours:

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}.$$

10 Dynamics of Rotation

10.1 Torque and Rotational Newton's Second Law

Torque

$$\tau = F\ell = rF\sin\phi = \mathbf{r} \times \mathbf{F} = F_{\tan}r$$

Equation of Motion

$$\sum \tau = I\alpha \Leftrightarrow \tau_{\text{net}} = I\alpha$$

ℓ, is the lever arm of $\vec{F}$
F is the magnitude of $\vec{F}$
r magnitude of $\vec{r}$ vector from the origin to where $\vec{F}$ acts
ϕ is the angle between $\vec{r}$ and $\vec{F}$

10.2 Work, Power, and Angular Momentum

Work done by a torque is computed as,

$$W = \int_{\theta_1}^{\theta_2} \tau d\theta$$

Work done by a constant torque,

$$W = \tau\Delta\theta$$

The total work done on a rotating body is equal to the work done by the net external torque and its computed as,

$$W_{\text{tot}} = \int_{\omega_1}^{\omega_2} I\omega d\omega = \frac{1}{2}I\omega_2^2 - \frac{1}{2}I\omega_1^2$$

The power due to torque on a rigid body is computed as,

$$P = \tau\omega$$

Angular momentum

The angular momentum of a rigid body rotating around a symmetry axis is computed as,

$$\vec{L} = I\vec{\omega}$$

The angular momentum of a particle relative to origin of an inertial frame of reference is computed as,

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$$

Where $\vec{r}$, is the position vector of particle relative to origin,
and $\vec{p}$, is the linear momentum of the particle.

Angular impulse–momentum theorem:

The sum of external torques on the system is computed as,

$$\sum \vec{\tau} = \frac{d\vec{L}}{dt}$$

10.3 Energy with Rotation and Translation

For rolling motion on a track with height change h:

$$\tfrac{1}{2}mv_{\text{CM}}^2 + \tfrac{1}{2}I\omega^2 + mgh = \text{constant}.$$

Conservation of angular momentum, when the net external torque acting on a system is zero, the total angular momentum of the system is constant (conserved)

10.4 The simple Pendulum

A simple pendulum is defined to have a point mass, also known as the pendulum bob, which is suspended from a string of length L with negligible mass.

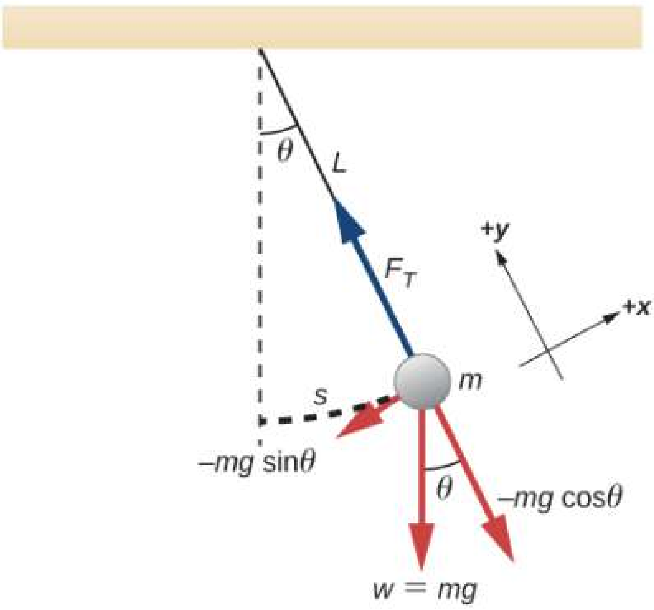


Figure 9: Simple Pendulum

Note. Although angular variables and torque are used, the pendulum bob is modeled as a point mass. Its kinetic energy is therefore $K = \frac{1}{2}mv^2$, not rotational kinetic energy.

Considering the torque on the pendulum, the force providing the restoring torque is the component of the weight of the pendulum bob that acts along the arc length.

$$\tau = -L(mg\sin(\theta)) \tag{2}$$
$$I\alpha = -L(mg\sin(\theta)) \tag{3}$$
$$I\frac{d^2\theta}{dt^2} = -L(mg\sin(\theta)) \tag{4}$$
$$mL^2\frac{d^2\theta}{dt^2} = -L(mg\sin(\theta)) \tag{5}$$
$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\sin(\theta) \tag{6}$$

For small angles (less than 15 degrees), $\sin\theta$ and θ differ by less than 1% so we can use the angle approximation $\sin\theta \approx \theta$,

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$$

The angular frequency is,

$$\omega = \sqrt{\frac{g}{L}}$$

and the period is,

$$T = 2\pi\sqrt{\frac{L}{g}}$$

In the case that the angle is bigger than 15 degrees, we can not ignore it anymore so the formula becomes:

$$T = 2\pi\sqrt{\frac{L}{g}}\cos\left(\frac{\theta_0}{2}\right)$$

10.5 Physical Pendulum

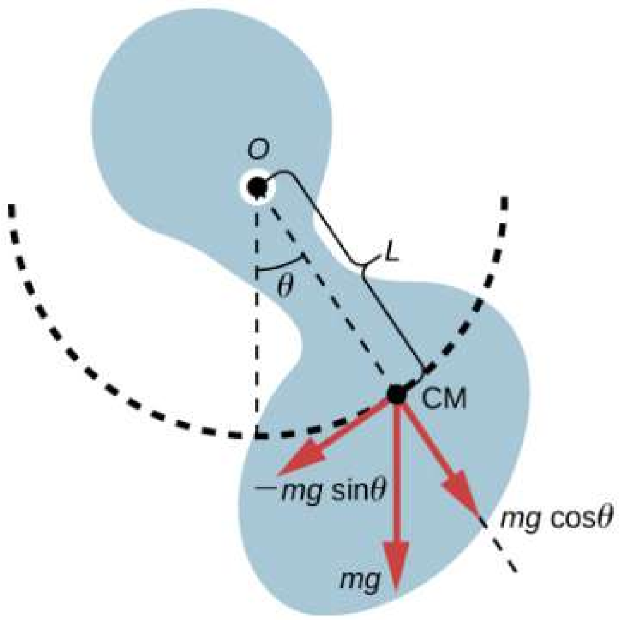


Figure 10: Physical pendulum

$$I\alpha = \tau_{\text{net}} = L(-mg)\sin\theta$$

Using the small angle approximation.

$$I_T\alpha = -L_{CM}(m_T g)\theta \tag{7}$$
$$I_T\frac{d^2\theta}{dt^2} = -L_{CM}(m_T g)\theta \tag{8}$$
$$\frac{d^2\theta}{dt^2} = -\left(\frac{m_T g L_{CM}}{I_T}\right)\theta \tag{9}$$

The solution is,

$$\theta(t) = \vartheta\cos(\omega t + \phi)$$

Where ϑ is the maximum angular displacement. The angular frequency,

$$\omega = \sqrt{\frac{m_T g L_{CM}}{I_T}}$$

The period is therefore,

$$T = 2\pi\sqrt{\frac{I_T}{m_T g L_{CM}}}$$

10.6 Linear ↔ Rotational Analogies

Linear	**Rotational**
x	θ
v	ω
a	α
m	I
F	τ
$p = mv$	$L = I\omega$
$W = F\,\Delta x$	$W = \tau\,\Delta\theta$

11 Thermodynamics

Thermodynamics links heat, work, and energy. It describes how energy is transferred between systems and how it determines macroscopic equilibrium.

11.1 Zeroth Law of Thermodynamics

If two systems are each in thermal equilibrium with a third system, they are in equilibrium with each other. This defines temperature as a measurable property and justifies the existence of thermometers.

11.2 Temperature Scales and Conversions

$$T_K = T_C + 273.15, \qquad T_F = \frac{9}{5}T_C + 32$$

The Celsius, Kelvin, and Fahrenheit scales are related linearly. Temperature measures average molecular kinetic energy.

11.3 Thermal Expansion

Material	Coefficient of Linear Expansion α (1/°C)	Coefficient of Volume Expansion β (1/°C)
Solids		
Aluminum	25×10^{-6}	75×10^{-6}
Brass	19×10^{-6}	56×10^{-6}
Copper	17×10^{-6}	51×10^{-6}
Gold	14×10^{-6}	42×10^{-6}
Iron or steel	12×10^{-6}	35×10^{-6}
Invar (nickel-iron alloy)	0.9×10^{-6}	2.7×10^{-6}
Lead	29×10^{-6}	87×10^{-6}
Silver	18×10^{-6}	54×10^{-6}
Glass (ordinary)	9×10^{-6}	27×10^{-6}
Glass (Pyrex®)	3×10^{-6}	9×10^{-6}
Quartz	0.4×10^{-6}	1×10^{-6}
Concrete, brick (approximate)	12×10^{-6}	36×10^{-6}
Marble (average)	2.5×10^{-6}	7.5×10^{-6}
Liquids		
Ether		1650×10^{-6}
Ethyl alcohol		1100×10^{-6}
Gasoline		950×10^{-6}

Figure 11: Table of coefficients of linear and volume expansion for various materials

Linear Expansion

$$\Delta L = \alpha L_0 \Delta T$$

Volume Expansion

$$\Delta V = \beta V_0 \Delta T, \quad \beta = 3\alpha$$

Area Expansion

$$\Delta A = 2\alpha A_0 \Delta T$$

Where α is denoted as the coefficient of linear expansion and β as the coefficients of volune expansion (both in $1/℃$

11.4 Heat and Specific Heat

Heat Q, in J, is defined as the product of the specific heat of what we are heating c, in $J/kg/K$, the mass of what we are heating m, in kg, and the change

Heat Transfer by Temperature Change

$$Q = mc\Delta T$$

Latent Heat of Phase Change

$$Q = mL_f, \text{ (melting/freezing)}$$
$$Q = mL_v, \text{ (vaporization/condensation)}$$

L — latent heat, c — specific heat capacity. 1 cal = 4.186 J.

The specific heat of ice is known to be,

$$c_{\text{ice}} = 2100\ J/kg \cdot K$$

The specific heat of liquid water is known to be,

$$c_{\text{water}} = 4184\ J/kg \cdot K$$

The specific heat of water vapor varies with temperature but is roughlt known to be,

$$c_{\text{Vapor}} = 1.9 - 2.0\ kJ/kg \cdot K$$

The latent of melting/freezing for water is known to be

$$L_f = 334\ kJ/kg$$

The latent of vaporization/condensation for water is known to be

$$L_v = 2260\ kJ/kg$$

11.5 Heat Transfer Mechanisms (conduction, convection, radiation)

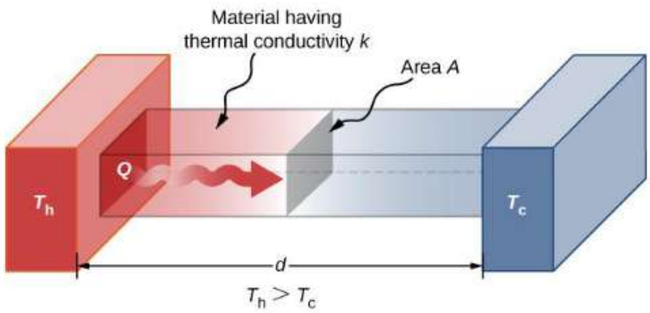


Figure 12: Conduction

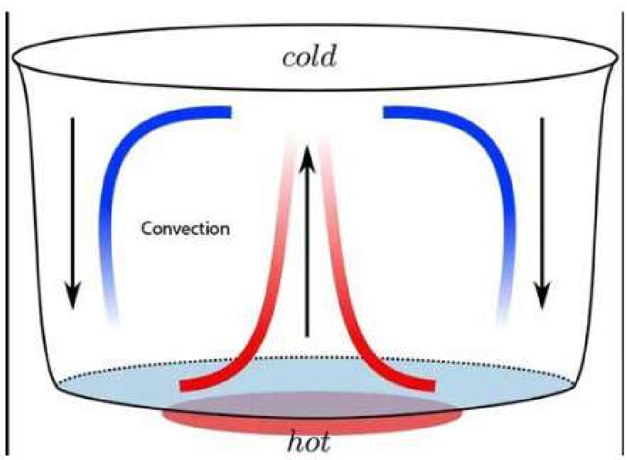


Figure 13: Convection

Conduction (rate of heat transfer)

$$P = \frac{dQ}{dt} = kA\frac{T_h - T_c}{L}$$

k is the coefficient of heat conduction of that material

Convection

$$P = hA(T_h - T_c)$$

Radiation (Stefan–Boltzmann Law)

$$P = \sigma e A T^4$$

Net heat transfer

$$P_{\text{net}} = \sigma e A(T_2^4 - T_1^4)$$

$\sigma = 5.67 \times 10^{-8}\ \mathrm{W/(m^2\,K^4)}$.

equivalence to the conduction equation

$$Q = P \cdot t$$

Composite conduction (series materials). For steady 1D conduction through multiple materials *in series* (same heat rate through each layer), use thermal resistances:

$$R_{\text{th,series}} = \sum_i R_i \qquad \text{with} \qquad R_i = \frac{L_i}{k_i A}.$$

Then the heat rate is

$$\dot{Q} = \frac{dQ}{dt} = \frac{\Delta T}{R_{\text{th,series}}}.$$

If the conducted heat causes a phase change (e.g. evaporation/condensation),

$$Q = \dot{Q}\,t, \qquad m = \frac{Q}{L},$$

where L is the latent heat of the phase change (J/kg).

Composite conduction (parallel paths). For conduction through multiple paths *in parallel* (same ΔT across each path), the equivalent thermal resistance satisfies:

$$\frac{1}{R_{\text{th,parallel}}} = \sum_i \frac{1}{R_i}, \qquad \text{with} \qquad R_i = \frac{L_i}{k_i A_i}.$$

Total heat rate:

$$\dot{Q}_{\text{tot}} = \frac{\Delta T}{R_{\text{th,parallel}}} = \sum_i \dot{Q}_i, \qquad \dot{Q}_i = \frac{\Delta T}{R_i}.$$

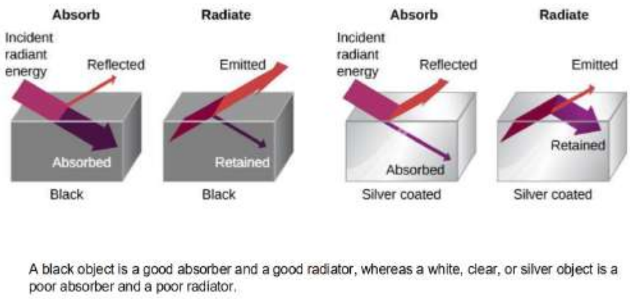


Figure 14: Different surfaces and their effects on radiation absorption/emission

11.6 Ideal Gas Law

$$pV = nRT = Nk_BT$$

where $R = 8.314$ J/(mol K) $= 0.0821$ L atm/(mol K) and $k_B = 1.38 \times 10^{-23}$ J/K. $N = N_A \cdot n$

11.7 The Van der Waals Equation of State

It is known that the ideal gas law is an approximation. It was improved by taking into account two factors. First, the attractive forces between molecules, which are stronger at higher density and reduce the pressure, are taken into by adding to the pressure a term equal to the square of the molar density multiplied by a positive coefficient a. Second, the volume of the molecules is represented by a positive constant b which can be thought of as the volume of a mole of molecules.

Van der Waals Equation of State

$$\left[p + a\left(\frac{n}{V}\right)^2\right](V - nb) = nRT$$

11.8 Kinetic Theory and Equipartition of Energy

The average kinetic energy of a molecule $\bar{K}$

$$\bar{K} = \frac{1}{2}m\bar{v}^2 = \frac{3}{2}k_BT$$

The root-mean-squared (rms) speed of a molecule

$$v_{\text{rms}} = \sqrt{\bar{v}^2} = \sqrt{\frac{3k_BT}{m}} = \sqrt{\frac{3RT}{M}}$$

Equipartition of energy each degree of freedom get $\frac{1}{2}k_BT$

$$E_{\text{int}} = \tfrac{3}{2}nRT$$

At constant volume a monotonic gas : $C_V = \frac{3}{2}R$, at constant pressure: $C_P = C_V + R$.

γ is defined as the ratio of the molar heat capacities,

$$\gamma = \frac{C_p}{C_V}$$

The ratio od the molar heat capacities for air is known as,

$\gamma \approx 1.4$

For a monatomic gas, $\gamma = \frac{C_P}{C_V} = \frac{5}{3}$.

The Molar heat capacity at constant volume for an ideal gas with d degrees of freedom is,

$$C_V = \frac{d}{2}R$$

Molar Heat Capacities of Dilute Ideal Gases at Room Temperature

Type of Molecule	Gas	C_p (J/mol K)	C_V (J/mol K)	$C_p - C_V$ (J/mol K)
Monatomic	Ideal	$\frac{5}{2}R = 20.79$	$\frac{3}{2}R = 12.47$	$R = 8.31$
Diatomic	Ideal	$\frac{7}{2}R = 29.10$	$\frac{5}{2}R = 20.79$	$R = 8.31$
Polyatomic	Ideal	$4R = 33.26$	$3R = 24.94$	$R = 8.31$

Table 3.3

Figure 15: Molar heat capacities of Dilute ideal gases at room temperature

11.9 First Law of Thermodynamics

Energy Conservation

$$\boxed{\Delta U = Q - W}$$

ΔU — change in internal energy, Q — heat added to system, W — work done by system. Positive Q adds energy; positive W means work *done by* the system.

Work at constant pressure: $W = p\Delta V$. For infinitesimal processes: $dU = \delta Q - \delta W$.

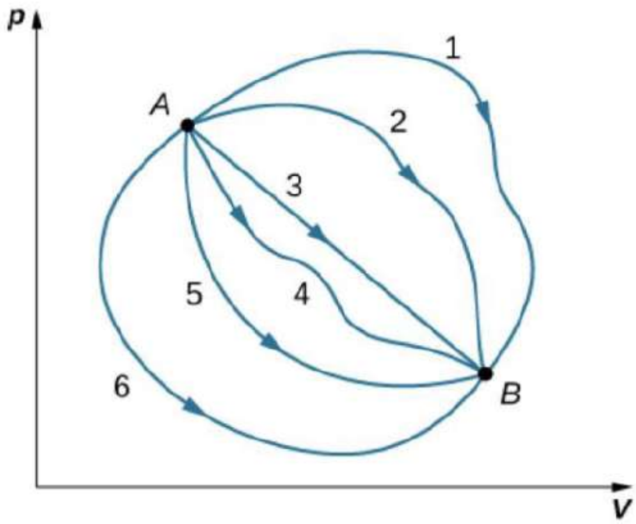


Figure 16: The path that it takes does does not affect it, only its initial and final conditions

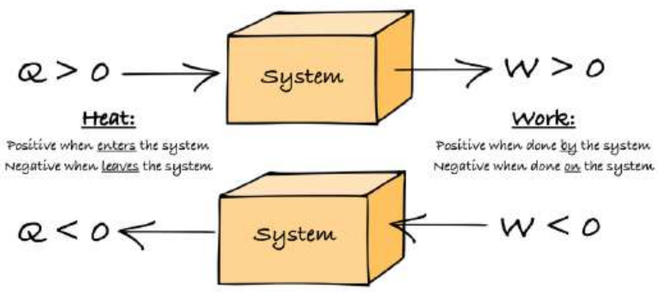


Figure 17: Be mindful of the sign

11.10 Enthalpy

Enthalpy combines internal energy and the work required to displace the surroundings:

$$H = U + pV$$

Differential form:

$$dH = dU + pdV + Vdp = dQ + Vdp$$

At constant pressure, $dH = dQ_p$, so the change in enthalpy equals the heat exchanged.

11.11 Thermodynamical systems

Open systems, Mass and energy can flow in or out

Closed system, mass is conserved but energy can flow in or out

Isolated systems, mass and energy are conserved

11.12 Thermodynamical variables

Thermodynamical variables (like volume, pressure, temperature, internal energy) can be **extensive** (properties that change with the mass, size, and extent of the system, like mass, volume or energy) and **intensive** (properties that do not depend on the size or amount of matter in the system, like temperature, pressure, or density).

A thermodynamical state can be associated to many microscopic states, but we don't need to know them all to know the thermodynamical variables.

The equation of state is satisfied at equilibrium $f(p, V, T) = 0$

11.13 Common Thermodynamic Processes

Isobaric Transformation

In an isobaric transformation pressure does not change,

$$\Delta P = 0 \rightarrow \Delta E_{tot} = Q - p\Delta V$$

In an ideal gas, $\frac{V_A}{T_A} = \frac{V_B}{T_B}$

$$W = p\Delta V$$

$$dQ = \underbrace{dE_{tot} + pdV}_{\Leftrightarrow dH} = nC_V dT + nRdT = n(C_V + R)T = nC_P dT$$

$dH0$ is the enthalpy and C_P is the molar heat capacity at constant pressure

Isochoric Transformation

In an isochoric transformation volume does not change,

$$\Delta V = 0 \rightarrow W = 0 \rightarrow Q = \Delta E_{tot} = nC_V dT = n(C_P - R)dT$$

In an ideal gas, $\frac{p_A}{T_A} = \frac{p_B}{T_B}$

Isothermal Transformation

In an isothermal transformation temperature does not change,

$$\Delta T = 0 \rightarrow \Delta E_{tot} = 0 \rightarrow Q = W$$

In an ideal gas, $p_A V_A = p_B V_B$ If its reversible,

$$\Delta W_{AB} = \int_A^B p dV = \int_A^B \frac{nRT}{V} dV = nRT \int_A^B \frac{dV}{V} = nRT \ln\left(\frac{V_B}{V_A}\right) = nRT \ln\left(\frac{p_A}{p_B}\right)$$

Isothermal expansion $W_{AB} > 0 \to Q_{AB} > 0$ the gas does work and absorbs heat.

Isothermal compression $W_{AB} < 0 \to Q_{AB} < 0$ work is done on the gas and heat is removed from the gas.

Adiabatic transformation

In an adiabatic transformation heat is not exchanged,

$$Q = 0 \to \Delta E_{tot} = -W = nC_V(T_B - T_A) = \frac{C_V}{R}(p_B V_B - p_A V_A) = \frac{1}{\gamma - 1}(p_b V_B - p_A V_A)$$

If its reversible,

$$dQ = 0 \to dE_{tot} + pdV = 0 \to nC_V dT + pdV = 0$$

For ideal gasses

$$d(PV) = nRdT \to W dp + pdV = nRdT$$

$$\to V dp + pdV = -\frac{R}{C_V} pdV$$

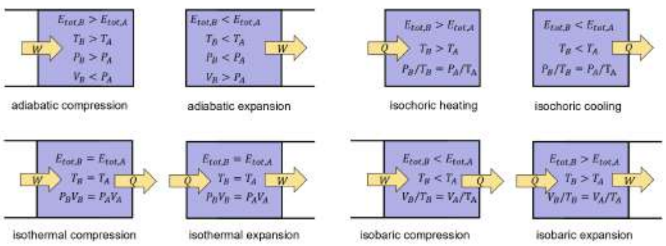
$$\to C_V dp + (C_V + R)pdV = 0$$

$$\to V dp + \frac{C_p}{C_V} pdV = 0 \to \frac{dp}{p} = -\gamma \frac{dV}{V}$$

$$\to \ln\left(\frac{p_B}{p_A}\right) = -\gamma \ln\left(\frac{V_B}{V_A}\right) \to \ln\left(\frac{p_B V_B^\gamma}{p_A V_A^\gamma}\right) = 0$$

$$\to PV^\gamma = \text{constant}$$

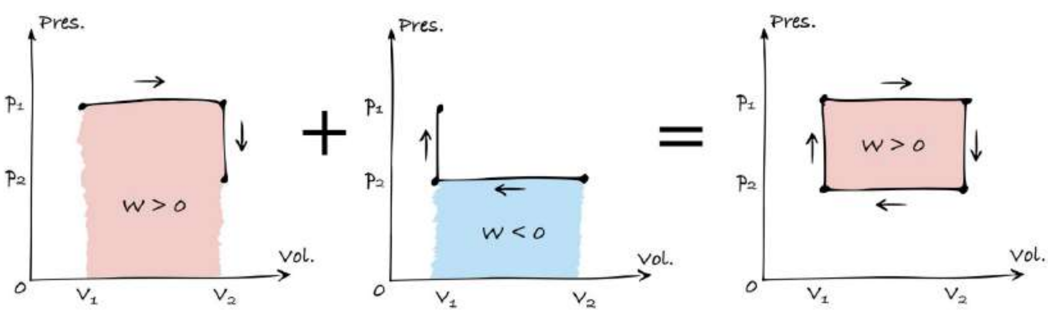
$$\to TV^{\gamma - 1} = \text{constant}$$



Cyclic transformation

In a cyclic transformation, we get back to the original point,

$$\Delta E_{tot} = 0 \to Q = W$$



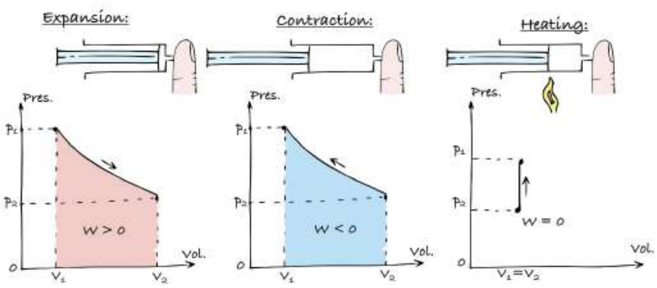


Figure 18: Work in a thermodynamical processes

Note! Work in a pressure volume graph is equal to the area under the curve.

11.14 Second Law of Thermodynamics

The second law sets the direction of thermodynamic processes.

Kelvin–Planck statement: It is impossible for any cyclic process to convert heat from a single reservoir completely into work.

Clausius statement: Heat never flows spontaneously from a colder to a hotter body.

Thermodynamic Sign Conventions for Heat and Work

Process	Convention
Heat added to system	$Q > 0$
Heat removed from system	$Q < 0$
Work done by system	$W > 0$
Work done on system	$W < 0$

Figure 19: Thermodynamic Sign Conventions for Heat and Work

11.15 Entropy

For a reversible cycle : $\Delta S = 0$

For a reversible process:

$$\Delta S = \int_A^B \frac{dQ_{\text{rev}}}{T}$$

For an irreversible process, $\Delta S_{\text{universe}} > 0$.
$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{environment}}$
Carnot engine:

$$\Delta S = \frac{Q_h}{T_h} - \frac{Q_c}{T_c}$$

Entropy as a state function (exam shortcut). The total entropy change depends only on the initial and final states, not on the path. For an ideal gas:

$$\Delta S = nC_V \ln\left(\frac{T_2}{T_1}\right) + nR\ln\left(\frac{V_2}{V_1}\right).$$

If $T_2 = T_1$, then

$$\Delta S = nR \ln\left(\frac{V_2}{V_1}\right).$$

Microscopic definition:

$$S = k_B \ln \Omega$$

where Ω is the number of accessible microstates.

11.16 Heat Engines and Carnot Cycle

Efficiency of a Heat Engine

$$e = \frac{W}{Q_h}$$

Efficiency of a Heat Engine in a closed loop

$$e = 1 - \frac{Q_c}{Q_h}$$

Carnot Efficiency (Ideal Limit)

$$e_{\text{Carnot}} = 1 - \frac{T_c}{T_h}$$

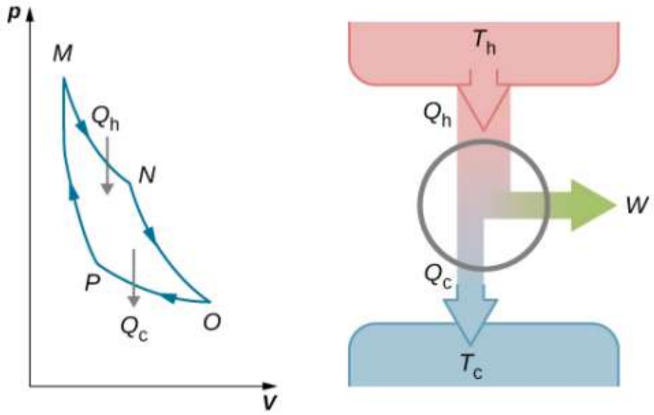


Figure 20: The total work done by the gas in the Carnot cycle is shown and given by the area enclosed by the loop MNOPM.

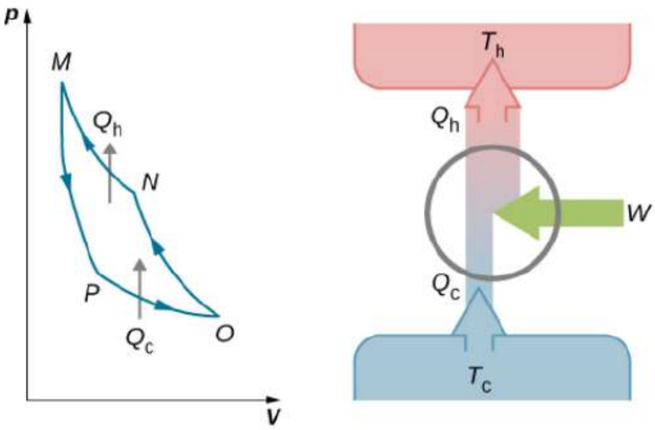


Figure 21: The work done on the gas in one cycle of the Carnot refrigerator is shown and given by the area enclosed by the loop MPONM.

Isothermal expansion

$$Q_h = nRTln(\frac{V_N}{V_M})$$

Adiabatic expansion

$$T_h V_N^{\gamma-1} = T_c V_O^{\gamma-1}$$

Isothermal compression

$$Q_h = W_{OP} = nRTln(\frac{V_O}{V_P})$$

Adiabatic compression

$$T_h V_P^{\gamma-1} = T_c V_M^{\gamma-1}$$

11.17 Refrigerators and Heat Pumps

$$K_R = \frac{Q_c}{W}, \qquad K_P = \frac{Q_h}{W}$$

For closed loops

$$K_R = \frac{Q_c}{Q_h - Q_c}, \qquad K_P = \frac{Q_h}{Q_h - Q_c}$$
$$W = Q_h - Q_c$$

For refrigerators Q_c is heat absorbed

K_P and K_R can also be referred as the coefficient of performance (COP)

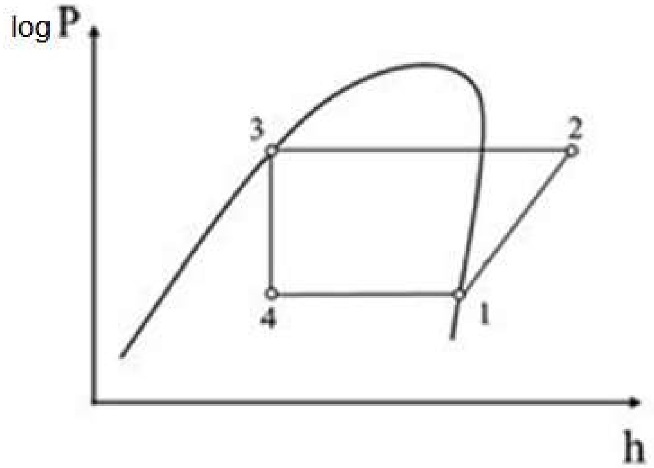


Figure 22: Pressure-Enthalpy (ph) diagram

$$K_P = COP = \frac{Q_h}{W}$$

W is the compressor work done in 1-2

Q_h is the heat delivered in 2-3

And the pumped heat from outside, Q_c, is the heat absorbed during 4-1

$$K_P = COP = \frac{Q_h}{W} = \frac{-\Delta h_{2-3}}{\Delta h_{1-2}}$$

For a reversible (Carnot) refrigerator:

$$K_{R,\text{Carnot}} = \frac{T_c}{T_h - T_c}$$

11.18 Reversible and Irreversible Processes

A reversible process is a process in which the system and environment can be restored to exactly the same initial state they were in before the process occurred (an example is ice cubes and liquid water)

An irreversible process is a process in which the system and its environment cannot be restored to their original states at the same time (an example is breaking an egg)

Implications for reversible processes

$$\Delta S = 0, \qquad \frac{Q_c}{T_c} = \frac{Q_c + W}{T_h}, \qquad W = Q_c(\frac{T_h}{T_c} - 1)$$

11.19 Third Law of Thermodynamics

As $T \to 0$ K, the entropy of a perfect crystal approaches zero:

$$\lim_{T \to 0} S = 0$$

11.20 Summary of the Four Laws of Thermodynamics

0th Law	Defines temperature via thermal equilibrium.
1st Law	Energy is conserved: $\Delta U = Q - W$.
2nd Law	Entropy of an isolated system never decreases.
3rd Law	$S \to 0$ as $T \to 0$ K.

12 Fluids and Gases

Fluids (liquids and gases) are substances that can flow and conform to the shape of their container. They exert forces perpendicular to any surface they contact.

12.1 Pressure in Fluids

Pressure is defined as force per unit area:

$$p = \frac{F_\perp}{A}$$

and is measured in pascals (N/m^2).

$$1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$$

12.2 Pressure and Depth

Pressure in a static fluid increases with depth to balance the weight of the fluid above:

$$g\rho A dy = dp \to \frac{dp}{dy} = -\rho g$$

Take y positive downward (depth) or flip signs if y is upward.

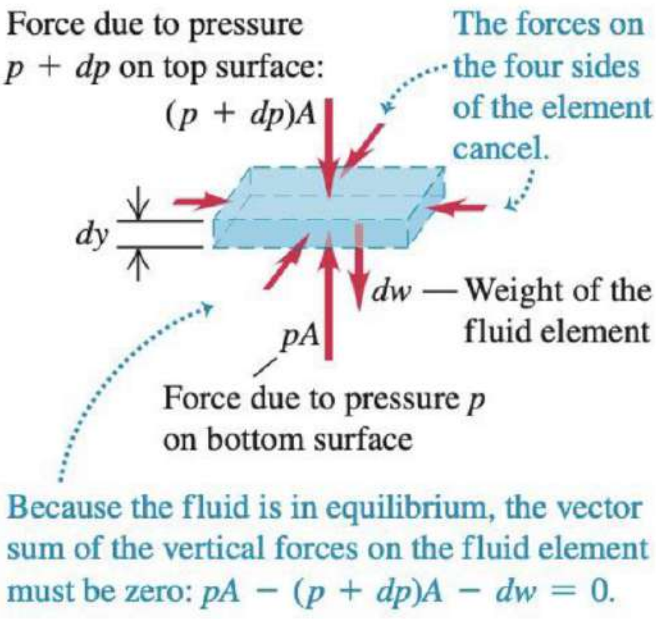


Figure 23: Pressure and depth

Integrating for constant density ρ:

$$p(y) = p_0 + \rho g y$$

Take y positive downward (depth) or flip signs if y is upward.

where p_0 is the pressure at the surface.

Gauge Pressure

$$p_g = \rho g h$$

Gauge pressure is the difference between absolute pressure and atmospheric pressure.

12.2.1 Constant relations

Keeping constant temperature T and number of moles N, implies,

$$pV = \text{constant}$$

Keeping constant pressure p and number of molecules N, implies,

$$\frac{V}{T} = \text{constant}$$

Keeping constant volume V and number of molecules N, implies,

$$\frac{P}{T} = \text{constant}$$

12.3 Pascal's Principle

A change in pressure applied to an enclosed fluid is transmitted undiminished to all points of the fluid and the walls of its container:

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

This is the operating principle behind hydraulic lifts and brakes.

12.4 Archimedes' Principle (Buoyancy)

An object submerged in a fluid experiences an upward buoyant force equal to the weight of the displaced fluid:

$$F_B = \rho_{\text{fluid}} V_{\text{disp}} g$$

If $F_B > W$, the object floats; if $F_B < W$, it sinks.

Apparent Weight

$$W_{\text{app}} = W - F_B$$

Condition for Floating

$$\rho_{\text{object}} = \rho_{\text{fluid}} \frac{V_{\text{disp}}}{V_{\text{object}}}$$

12.5 Continuity Equation (Conservation of Mass)

For incompressible fluid flow:

$$A_1 v_1 = A_2 v_2$$

This means that when the cross-sectional area decreases, velocity increases.

12.6 Bernoulli's Equation (Conservation of Energy in a Fluid)

Derived from the work–energy theorem for flowing fluids:

$$p + \frac{1}{2}\rho v^2 + \rho g y = \text{constant}$$

along a streamline.

- p — pressure energy per unit volume
- $\frac{1}{2}\rho v^2$ — kinetic energy per unit volume
- $\rho g y$ — potential energy per unit volume

12.7 Applications of Bernoulli's Principle

- **Venturi effect:** pressure decreases where velocity increases.
- **Lift on airplane wings:** faster air over the top of the wing reduces pressure, generating lift.
- **Flow speed from a tank:** using Bernoulli and $y_1 - y_2 = h$:

  $$v = \sqrt{2gh}$$

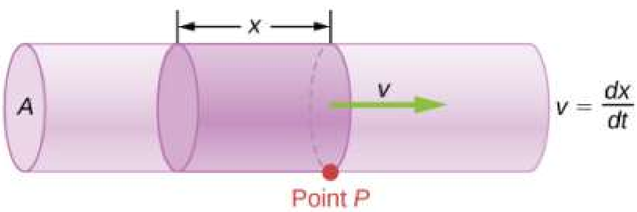
  (Torricelli's theorem)

12.8 Fluid Dynamics

The volume of fluid passing by a given location through an area during a period of time is called flow rate Q, or more precisely, volume flow rate. It is written as,

$$Q = \frac{dV}{dt}$$

Where V is the volume and t is the time elapsed.



The volume of the cylinder above is Ax, so the flow rate is,

Flow rate

$$Q = \frac{dV}{dt} = \frac{d}{dt}(Ax) = A\frac{dx}{dt} = Av$$

Where A is the cross-sectional area and v is the magnitude of the velocity.

12.9 Viscosity and Laminar Flow

For laminar flow of a viscous fluid in a cylindrical pipe (Poiseuille's law):

$$Q = \frac{\pi R^4 (p_1 - p_2)}{8\eta L}$$

where Q is the volume flow rate, η is the dynamic viscosity, R is the radius of the pipe, p_1 and p_2 are the initial and final pressures of the liquid in the pipe, and L is the pipe length.

12.10 Maxwell-Boltzmann distributions of speeds

The distribution function for speeds of particles in an ideal gas at temperature T is,

$$f(v) = \frac{4}{\sqrt{\pi}} \left(\frac{m}{2k_b T}\right)^{3/2} v^2 e^{\left(-mv^2/(2k_B T)\right)}$$

Where m is the mass of one particle T is the temperature and v is the velocity.

12.11 Dalton's law

When the volume and temperature are kept constant, when we combine the gases in general, the total pressure can be found using,

$$p_{total} = \sum_{i=1}^{n} p_i$$

12.12 Summary of Key Fluid Relations

$$\begin{aligned} p &= \frac{F_\perp}{A} \\ p(y) &= p_0 + \rho g y \\ F_B &= \rho V g \\ A_1 v_1 &= A_2 v_2 \\ p + \frac{1}{2}\rho v^2 + \rho g y &= \text{constant} \end{aligned}$$

13 Gravitation

Gravitation is the weakest but most universal fundamental interaction, acting between all masses and shaping astronomical motion.

Gravitation is the mutual attraction between masses. The SI unit of force is the newton (N); the gravitational constant G is a universal proportionality constant.

13.1 Newton's Law of Universal Gravitation

Gravitational Force Between Two Point Masses

$$\mathbf{F}_{12} = -G\frac{m_1 m_2}{r^2}\,\hat{\mathbf{r}}_{12}$$

Where:

- $G = 6.67 \times 10^{-11}\ \mathrm{N\,m^2/kg^2}$
- m_1, m_2 are the interacting masses
- r is the distance between them
- $\hat{\mathbf{r}}_{12}$ is the unit vector from mass 1 to mass 2

13.2 Superposition of Gravitational Forces

For multiple sources, the net gravitational force is the vector sum:

$$\mathbf{F}_{\text{net}} = \sum_i \mathbf{F}_i$$

13.3 Gravitational Field and Acceleration

The gravitational field (acceleration due to gravity) created by a mass M is:

$$\mathbf{g} = -G\frac{M}{r^2}\,\hat{\mathbf{r}}$$

At Earth's surface:

$$g = 9.81\ \mathrm{m/s^2}$$

At altitude h above the surface ($r = R_E + h$):

$$g_h = G\frac{M_E}{(R_E + h)^2}$$

13.4 Cavenish Experiment (Measuring G)

Henry Cavendish (1798) measured G using a torsion balance with small lead spheres. This experiment also allowed determination of Earth's mass:

$$M_E = \frac{gR_E^2}{G} \approx 5.97 \times 10^{24}\ \mathrm{kg}$$

13.5 Gravitational Potential Energy

Work done by gravity from r_1 to r_2:

$$W = \int_{r_1}^{r_2} F\,dr = GMm\int_{r_1}^{r_2} \frac{dr}{r^2}$$

Potential Energy Function

$$U(r) = -G\frac{Mm}{r}$$

Near Earth's surface, $U = mgh$ is an approximation of this general formula.

13.6 Gravitational Potential

Potential energy per unit mass:

$$\Phi(r) = \frac{U}{m} = -G\frac{M}{r}$$

The gradient of the potential gives the gravitational field:

$$\mathbf{g} = -\boldsymbol{\nabla}\Phi$$

13.7 Gravitational Field Inside Earth

Assuming uniform density:

$$g(r) = G\frac{M(r)}{r^2} = G\frac{(4/3)\pi\rho r^3}{r^2} = \frac{4}{3}\pi G\rho r$$

Thus, inside a uniform Earth, $g(r)$ increases linearly with the distance from 0 at $r = 0$ to a maximum at the surface.

13.8 Escape Velocity

To escape Earth's gravitational field:

$$\frac{1}{2}mv_{\text{esc}}^2 = G\frac{M_E m}{R_E}$$

Escape Speed

$$v_{\text{esc}} = \sqrt{\frac{2GM_E}{R_E}} \approx 11.2 \text{ km/s}$$

13.9 Satellites and Circular Orbits

For a satellite of mass m orbiting Earth:

$$\frac{GM_E m}{r^2} = \frac{mv^2}{r}$$

Hence:

$$v_{\text{orbit}} = \sqrt{\frac{GM_E}{r}}$$

Orbital period:

$$T = 2\pi\sqrt{\frac{r^3}{GM_E}}$$

At altitude $h = 400$ km (International Space Station):

$$v_{\text{orbit}} \approx 7.67 \text{ km/s}, \quad T \approx 92 \text{ min}$$

For a body of mass m orbitting another body of mass M we have that,

$$v_{\text{orbit}} = \sqrt{\frac{GM}{r}}$$

Orbital period:

$$T = 2\pi\sqrt{\frac{r^3}{GM}}$$

13.10 Energy in Orbits

Total mechanical energy:

$$E = K + U = -\frac{GMm}{2r}$$

Negative energy indicates a bound (elliptical) orbit.

13.11 Kepler's Laws of Planetary Motion

1. **First Law (Elliptical Orbits)** — Planets move in ellipses with the Sun at one focus:
$$r(\theta) = \frac{\alpha}{1 + e\cos\theta}$$
2. **Second Law (Equal Areas)** — A line joining a planet and the Sun sweeps out equal areas in equal times:
$$\frac{dA}{dt} = \frac{L}{2m} = \text{constant}$$
3. **Third Law (Harmonic Law)** — The square of the orbital period is proportional to the cube of the semi-major axis:
$$T^2 = \frac{4\pi^2}{GM}a^3$$

If we are given a body b_1 orbits another bode b_2 and we are told we do not approximate the orbit to be circular (i.e. we keep it elliptical) we know that the body b_2 will be one of the focuses of the elliptical orbit and the tangential force that is applied on b_1 will be largest when it is closest to b_1.

13.12 Einstein's View of Gravitation (General Relativity)

In Einstein's theory, gravity is not a force but a manifestation of spacetime curvature produced by mass and energy. Massive objects curve spacetime, and smaller objects follow geodesics in this curved geometry.

14 Electrostatics

Electrostatics deals with forces, fields, and potentials produced by stationary electric charges. The SI unit of charge is the **coulomb (C)**.

$$1 \text{ C} = 6.242 \times 10^{18} \text{ electrons}$$

14.1 Conductors, Insulators, and Polarization

Conductors contain mobile electrons that move freely in response to an electric field; insulators have tightly bound charges. When a charged object is brought near a neutral one, molecular or electronic polarization separates positive and negative charge centers, producing attraction without net charge transfer.

14.2 Coulomb's Law

Force Between Two Point Charges

$$\mathbf{F}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}\hat{\mathbf{r}}_{12} \qquad |\mathbf{F}_{12}| = \frac{1}{4\pi\varepsilon}\frac{|q_1q_2|}{r_{12}^2}$$

$$\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$$

Direction:

- Force is **repulsive** for like charges ($q_1 q_2 > 0$).
- Force is **attractive** for unlike charges ($q_1 q_2 < 0$).

14.3 Principle of Superposition

For multiple charges:

$$\mathbf{F}_{\text{net}} = \sum_i \mathbf{F}_i = \frac{1}{4\pi\varepsilon_0} \sum_i \frac{q q_i}{r_i^2} \hat{\mathbf{r}}_i$$

Each force acts along the line joining the charges, and vector addition determines the net result.

14.4 Electric Field

Electric field is the force per unit charge experienced by a test charge q_0:

$$\mathbf{E} = \frac{\mathbf{F}}{q_0}$$

Electric Field of a Point Charge

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

- Field lines point away from positive charges and toward negative charges.
- The field is tangent to the lines at any point.
- The density of lines indicates field strength.

14.5 Electric Field of Multiple Point Charges

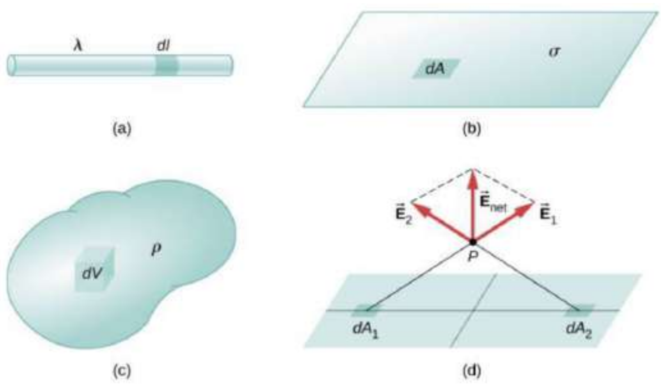
For several point charges, the total electric field at a point P is the vector sum of the individual fields from each charge:

$$\vec{E}_{\text{total}} = \sum_i \vec{E}_i = \frac{1}{4\pi\varepsilon_0} \sum_i \frac{q_i}{r_i^2} \hat{r}_i$$

Each $\vec{E}_i$ points along the line joining q_i to the observation point P.

This expresses the **principle of superposition**: electric fields add vectorially, and each charge acts independently of the others.

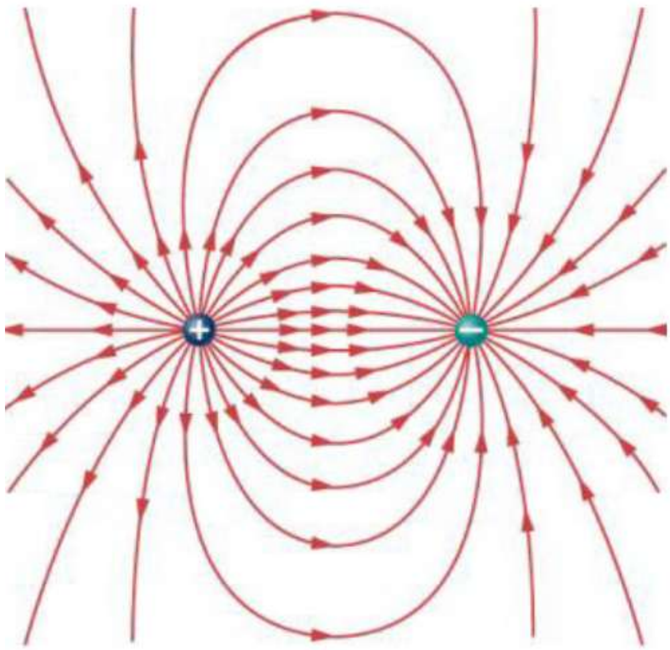
14.6 Electric Field of Continuous Charge Distributions



- **Line of charge:** $dq = \lambda\, dl$
- **Surface charge:** $dq = \sigma\, dA$
- **Volume charge:** $dq = \rho\, dV$

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0} \int \frac{dq}{r^2}\, \hat{\mathbf{r}}$$

14.7 Electric Field from a Dipole



For an electric dipole ($+q$, $-q$ separated by distance $2a$):

$$\mathbf{E}_{\text{axial}} = \frac{1}{4\pi\varepsilon_0}\frac{2p\cos\theta}{r^3}, \quad \mathbf{E}_{\text{equatorial}} = \frac{1}{4\pi\varepsilon_0}\frac{p\sin\theta}{r^3}$$

where the dipole moment $\mathbf{p} = q(2a)\, \hat{\mathbf{n}}$.

$$\vec{E}_{\text{dip}} = \frac{1}{4\pi\epsilon_0}\frac{1}{r^3}\left[3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}\right]$$

14.8 Electric Flux

Electric flux through a surface is the number of field lines passing through it:

$$\Phi_E = \int \mathbf{E}\cdot d\mathbf{A} = E_\perp A$$

Units: $\mathrm{N\,m^2/C}$.

14.9 Gauss's Law

Integral Form

$$\oint_S \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0}$$

Applies to symmetric charge distributions (spherical, cylindrical, planar).

- **Spherical symmetry.** Applies when the charge distribution is invariant under rotations (e.g. point charge, uniformly charged sphere or spherical shell). The electric field depends only on the distance r from the center and points radially:

$$\vec{E}(r) = E(r)\,\hat{r}, \qquad E(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q_{\text{enc}}}{r^2}.$$

  Outside a spherically symmetric charge distribution, the field is identical to that of a point charge with total charge Q_{enc}. Inside, Q_{enc} is the charge enclosed within radius r.

- **Cylindrical symmetry.** Applies when the charge distribution is invariant under translations along one axis and rotations around that axis (e.g. infinite line charge, long uniformly charged cylinder). The electric field depends only on the radial distance r from the axis and points radially outward:

$$\vec{E}(r) = E(r)\,\hat{r}, \qquad E(r) = \frac{\lambda}{2\pi\varepsilon_0 r},$$

  where λ is the charge per unit length enclosed by the Gaussian surface.

- **Planar symmetry.** Applies when the charge distribution is invariant under translations parallel to a plane (e.g. infinite charged sheet, two parallel plates). The electric field is uniform, perpendicular to the plane, and independent of the distance from the plane:

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\,\hat{n}.$$

  For two parallel plates with surface charge densities $\pm\sigma$, the fields add between the plates and cancel outside, giving

$$E = \frac{\sigma}{\varepsilon_0}.$$

14.10 Electric Potential Energy

The electric potential energy is derived from:

$$U_P = -\int_R^P \vec{F} \cdot d\vec{l}$$

Adding the explicit expressions, the work required to move a charge in an electric field:

$$U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$$

For multiple charges:

$$U_{\text{total}} = \frac{1}{4\pi\varepsilon_0}\sum_{i<j}\frac{q_i q_j}{r_{ij}}$$

14.11 Electric Potential (Voltage)

The electric potential is derived from:

$$V_P = -\int_R^P \vec{E} \cdot d\vec{l}$$

Potential at a point in an electric field:

$$V = \frac{U}{q} = -\int \mathbf{E} \cdot d\mathbf{r}$$

For a point charge:

$$V = \frac{1}{4\pi\varepsilon_0} \frac{q}{r}$$

Relation between potential and field:

$$\mathbf{E} = -\boldsymbol{\nabla} V$$

Electric potential decreases in the direction of the electric field.

Equipotential surfaces are perpendicular to **E**; no work is required to move a charge along them.

14.12 Relationship between U_P and V_P

$$U_P = qV_P$$
$$\vec{E} = \frac{\vec{F}}{q}$$

14.13 Potential Difference and Work

The potential difference (AKA work done) between two points is derived from:

$$W_{12} = \int_{P_1}^{P_2} \vec{F} \cdot d\vec{l}$$

$$W = -\Delta U = -q\Delta V \qquad \text{or} \qquad \Delta V = \frac{\Delta U}{q}$$

The potential is measured in:

$$1\text{V} = 1\text{J/C}$$

Note the convention:

$$-\Delta V = -(V_B - V_A) = V_A - V_B = -\int_A^B \mathbf{E} \cdot d\mathbf{r}$$

Positive work is done by the electric field when a positive charge moves from a higher to a lower potential.

Using the work integral in practice The general definition of work is

$$W = \int \vec{F} \cdot d\vec{r}.$$

If the force is constant and has a fixed direction, the integral reduces to the displacement along the force direction:

$$W = \int \vec{F} \cdot d\vec{r} = F \int d\ell_{\parallel} = F\,\Delta\ell_{\parallel}.$$

For a uniform electric field $\vec{E} = E\,\hat{y}$,

$$W = q \int \vec{E} \cdot d\vec{r} = qE \int dy = qE\,\Delta y,$$

independent of the actual path taken.

14.14 Electric Potential of Point and Distributed Charges

For a single point charge:

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

and for several charges, potential adds as a scalar:

$$V_{\text{total}} = \sum_i \frac{1}{4\pi\varepsilon_0}\frac{q_i}{r_i}$$

For continuous charge distributions:

$$V = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r}.$$

Equipotential surfaces are always perpendicular to $\mathbf{E}$.

14.15 Capacitance (Optional Intro)

Capacitance is the ability to store charge per potential difference:

$$C = \frac{Q}{V}$$

For a parallel-plate capacitor:

$$C = \varepsilon_0 \frac{A}{d}$$

Energy stored:

$$U = \frac{1}{2}CV^2$$

14.16 Summary of Electrostatic Quantities

$$\begin{aligned}
\mathbf{F} &= \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\,\hat{\mathbf{r}} \\
\mathbf{E} &= \frac{\mathbf{F}}{q} \\
V &= \frac{U}{q} = -\int \mathbf{E}\cdot d\mathbf{r} \\
U &= qV \\
\oint \mathbf{E}\cdot d\mathbf{A} &= \frac{Q_{\text{enclosed}}}{\varepsilon_0}
\end{aligned}$$

15 Magnetic Forces and Fields

Magnetism arises from moving charges and intrinsic magnetic dipoles. A magnetic field forms closed loops: field lines emerge from a magnet's north pole and enter the south pole, continuing through the magnet's interior.

Magnetic fields exert forces on moving charges and current-carrying wires. The SI unit of magnetic field strength is the **tesla (T)**:

$$1\ \text{T} = 1\ \text{N}/(\text{A}\,\text{m})$$

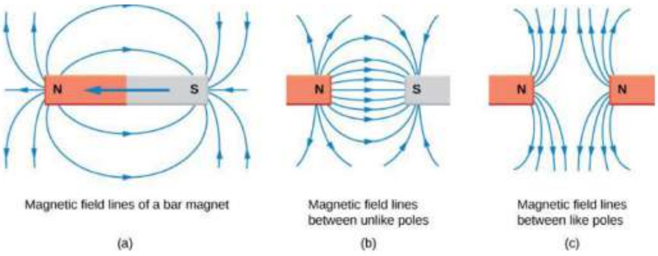


Figure 24: Magnetic field lines

15.1 Magnetic Forces on Charges and Currents

Magnetic Field and Lorentz Force

The force on a charge q moving with velocity $\mathbf{v}$ in an electric and magnetic field is:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

The purely magnetic part is the **Lorentz force**:

$$\boxed{\mathbf{F}_B = q\,\mathbf{v} \times \mathbf{B}}$$

- $\mathbf{F}_B$ is always perpendicular to both $\mathbf{v}$ and $\mathbf{B}$.
- No work is done by $\mathbf{F}_B$ (it changes direction but not speed).

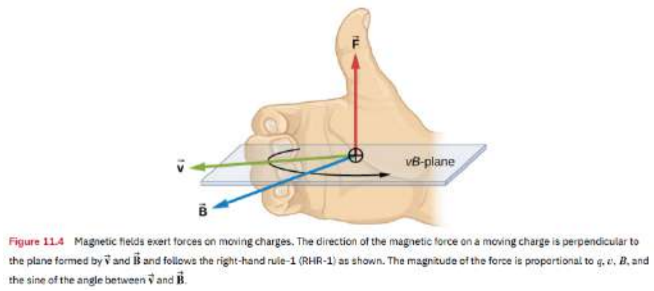


Figure 25: Right-Hand rule

Note! for negatively charged particles flip the force's direction.

Magnitude of the Magnetic Force

$$F = qvB\sin\theta$$

Motion of a Charged Particle in a Uniform Magnetic Field

When $\mathbf{v} \perp \mathbf{B}$, the charge follows a circular path.

Radius of Path

$$r = \frac{mv}{qB}$$

Period of Motion

$$T = \frac{2\pi m}{qB}$$

Frequency

$$f = \frac{1}{T} = \frac{qB}{2\pi m}$$

Helical Motion (general case)

$$v_{\parallel} = v\cos\theta, \quad v_{\perp} = v\sin\theta$$

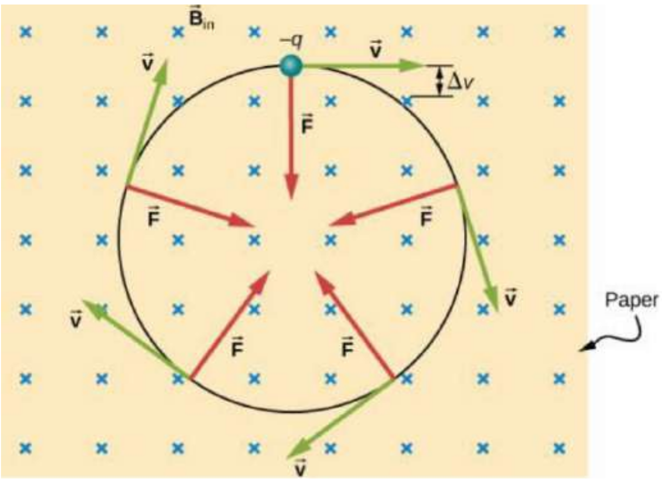


Figure 26: Motion of a charged particle in a uniform magnetic field

A negatively charged particle moves in the plane of the paper in a region where the magnetic field is perpendicular to the paper (represented by the small ×'s like the tails of arrows). The magnetic force is perpendicular to the velocity, so the velocity changes in direction, but not magnitude. The result is uniform circular motion. (Note that because the charge is negative, the force is opposite in direction to the prediction of the right-hand rule.)

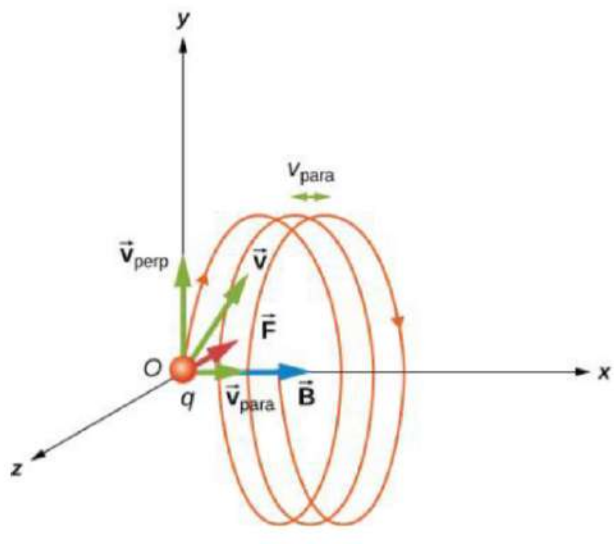


Figure 27: Motion of a charged particle in a non-perpendicular field

Magnetic Force on a Current-Carrying Wire

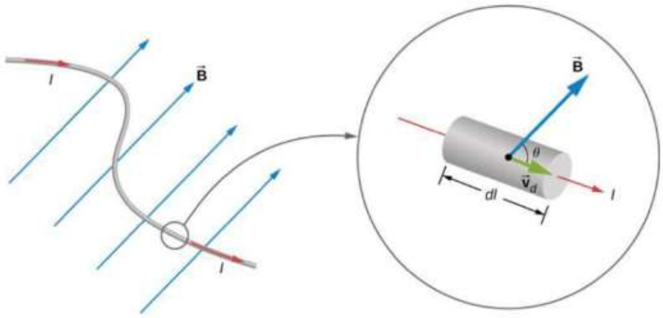


Figure 28: Magnetic force on a current-carrying wire

A current-carrying conductor in a magnetic field experiences a force given by:

$$\mathbf{F} = I\,\boldsymbol{\ell} \times \mathbf{B}$$

For a straight wire of length L:

$$F = ILB\sin\theta$$

The direction is given by the **right-hand rule**: $\rightarrow$ force ($\mathbf{F}$).
thumb $\rightarrow$ current (I), fingers $\rightarrow$ field ($\mathbf{B}$), palm

Torque on a Current Loop

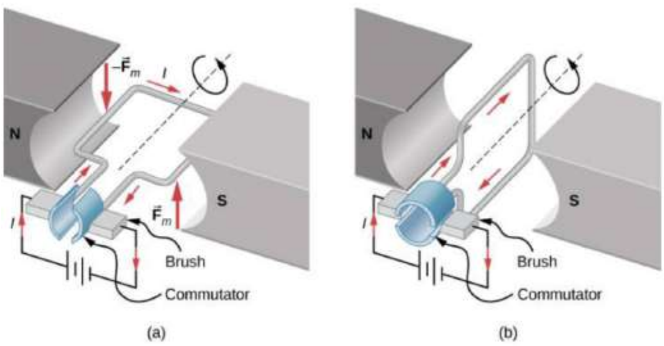


Figure 29: Force and torque on a current loop

A current loop in a magnetic field experiences a torque that tends to align its magnetic dipole moment μ with $\mathbf{B}$:

$$\tau = \mu \times \mathbf{B}$$

Magnetic Dipole Moment

$$\mu = NIA\,\hat{\mathbf{n}}$$

where:

- N — number of turns,
- I — current,
- A — area of loop,

- $\hat{\mathbf{n}}$ — unit normal vector to the plane of the loop.

The potential energy of a magnetic dipole in a field is

$$U = -\mu \cdot \mathbf{B}$$

15.2 Sources of Magnetic Fields

Biot–Savart Law (General Form)

A current element $I\,d\vec{\ell}$ produces a magnetic field at a point P given by the Biot–Savart law:

Biot–Savart law

$$\vec{B} = \frac{\mu_0}{4\pi} \int_{\text{wire}} \frac{I\,d\vec{\ell} \times \hat{r}}{r^2}$$

Biot–Savart law in differential form

$$d\vec{B} = \frac{\mu_0}{4\pi} I \frac{d\vec{l} \times \hat{r}}{r^2}$$

Biot–Savart law (magnitude)

$$dB = \frac{\mu_0}{4\pi} I \frac{dl \sin(\theta)}{r^2}$$

- dB is the magnetic field from a small current element.
- μ_0 is the permeability of free space $4\pi \times 10^{-7}\,\mathrm{T \cdot m/A}$.
- I is the current.
- dl is the length of the current element.
- $\hat{r}$ is the unit vector pointing from the current element to the point of interest.
- r is the distance from the current element to the point of interest.
- θ is the angle between the current direction dl and the vector $\vec{r}$.

Magnetic Field of a Long Straight Current

From the Biot–Savart law or Ampère's law, the magnetic field of an infinitely long straight current is

$$B = \frac{\mu_0 I}{2\pi r}$$

where $\mu_0 = 4\pi \times 10^{-7}$ N/A^2 is the permeability of free space and r is the distance from the wire.

Magnetic field due to a thin straight wire A section of thin, straight current-carrying wire. The independent variable θ has the limits θ_1 and θ_2.

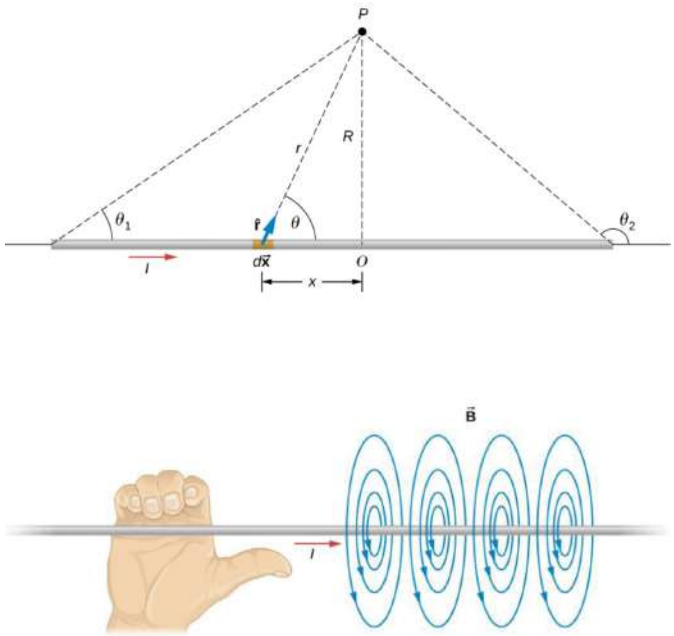
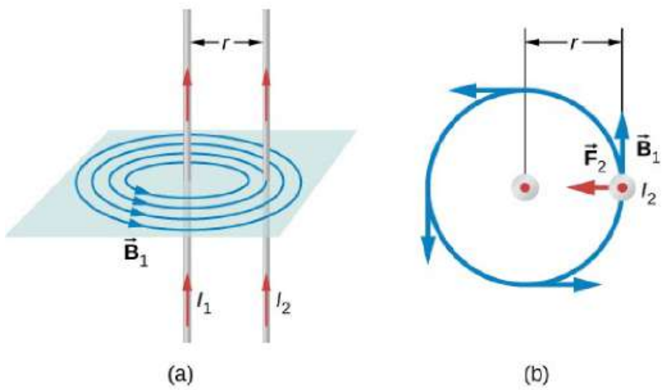


Figure 30: Magnetic field due to a thin straight wire

Magnetic field due to a long straight wire

$$B = \frac{\mu_0 I}{2\pi R}$$

Magnetic Force Between Two Parallel Currents



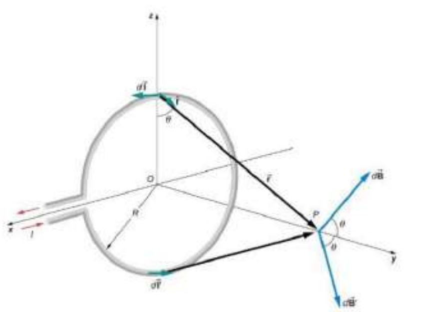
(a) The magnetic field produced by a long straight conductor is perpendicular to a parallel conductor, as indicated by the right-hand rule.

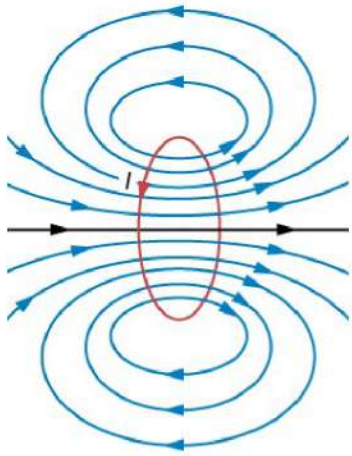
(b) A view from above of the two wires shown in (a), with one magnetic field shown for wire 1. The right-hand rule shows that the force between the parallel conductors is attractive when the currents are in the same direction. A similar analysis shows that the force is repulsive between currents in opposite directions.

Force between two parallel currents

$$\frac{F}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi r}$$

Magnetic Field from a Current Loop





For the initial conditions,

$$d\vec{l} \parallel y\text{-axis} \qquad \underbrace{\hat{r} \perp d\vec{l}}_{\text{in } zy\text{-plane}}$$

where the unit vector $\hat{\jmath}$ is parallel to the y-axis.

We can compute the magnetic field as

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{I dl \sin(\pi/2)}{r^2} = \frac{\mu_0}{4\pi} \frac{I dl}{y^2 + R^2}$$

$$\vec{B} = \hat{\jmath} \frac{\mu_0}{4\pi} \frac{IR}{(y^2 + R^2)^{3/2}} \int_{\text{loop}}^{2\pi R} dl \;\Rightarrow\; \vec{B} = \frac{\mu_0}{2} \frac{IR^2}{(y^2 + R^2)^{3/2}} \hat{\jmath}$$

Since we know the dipole moment,

$$\vec{\mu} = IA\hat{n} \qquad \text{(dipole moment)}$$

we can reformulate it as

$$\vec{B} = \frac{\mu_0 \mu \hat{\jmath}}{2\pi(y^2 + R^2)^{3/2}}.$$

Magnetic field due to a current loop (on axis)

$$\vec{B} = \frac{\mu_0}{2} \frac{IR^2}{(y^2 + R^2)^{3/2}} \hat{\jmath} \qquad \vec{B} = \frac{\mu_0 \mu \hat{\jmath}}{2\pi(y^2 + R^2)^{3/2}}$$

In the special case where $y = 0$ (point P at the center of the loop):

Field at the center of a circular loop

$$\vec{B} = \frac{\mu_0 I}{2R} \hat{\jmath}$$

Field at the center of a flat coil with n turns

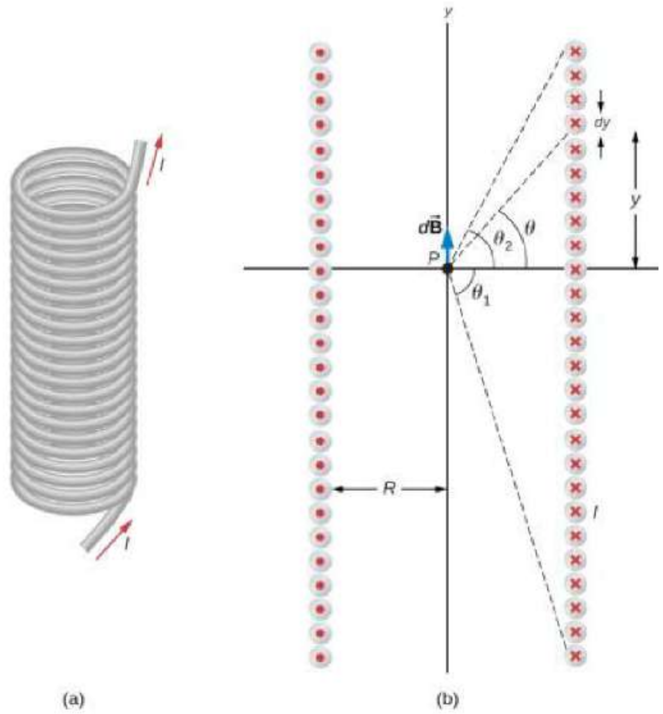
$$B = \frac{\mu_0 n I}{2R}$$

In the special case where $y \gg R$:

Far-field approximation (magnetic dipole)

$$\vec{B} = \frac{\mu_0 \vec{\mu}}{2\pi y^3}$$

Solenoids and Toroids



(a) A solenoid is a long wire wound in the shape of a helix.
(b) The magnetic field at the point P on the axis of the solenoid is the net field due to all of the current loops.

Solenoid:

$$\text{turns: } N, \quad \text{length: } L, \quad \text{current: } I, \quad \text{turns per length: } n = \frac{N}{L}$$

$$dI = \frac{NI}{L} dy$$

In the specific case where $L \gg R$:

$$\vec{B}_{\text{inside}} = \frac{\mu_0 IN}{L} = \mu_0 n I \hat{\jmath}, \qquad \vec{B}_{\text{outside}} \approx 0$$

Magnetic field strength inside a solenoid

$$B = \mu_0 n I$$

Magnetic field strength inside a toroid

$$B = \frac{\mu_0 N I}{2\pi r}$$

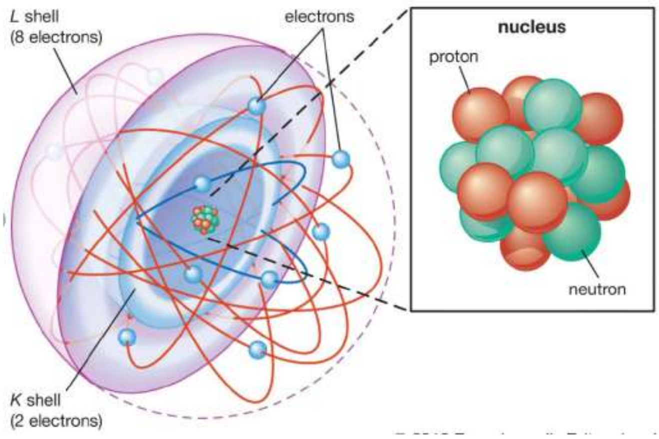
Ampère's Law

Ampère's law (integral form)

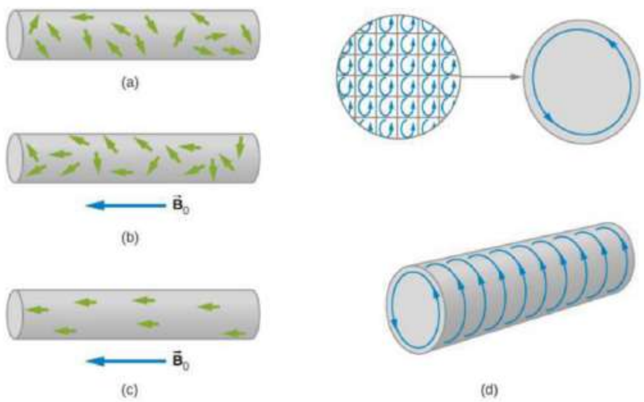
$$\oint \mathbf{B} \cdot d\ell = \mu_0 I_{\text{enclosed}}$$

This relates the circulation of **B** around a closed loop to the total current passing through the loop and is particularly useful for long straight wires, solenoids, and toroids.

15.3 Magnetism in Matter



Paramagnetism



The alignment process in paramagnetic material filling a solenoid (not shown):
(a) Without an applied field, the magnetic dipoles are randomly oriented.
(b) With a field, partial alignment occurs.
(c) An equivalent representation of part (b).
(d) The internal currents cancel, leaving an effective surface current that produces a magnetic field similar to that of a finite solenoid.

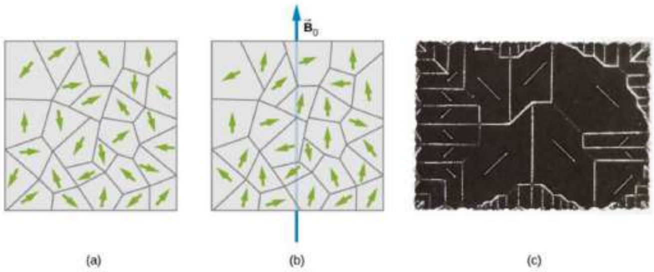
Magnetic permeability

$$\mu = (1 + \chi)\mu_0$$

Field of a solenoid filled with paramagnetic material

$$B = \mu n I$$

Ferromagnetism



(a) Domains are randomly oriented in an unmagnetized ferromagnetic sample such as iron. The arrows represent the orientations of the magnetic dipoles within the domains.
(b) In an applied field, domains tend to align, leading to strong net magnetization.

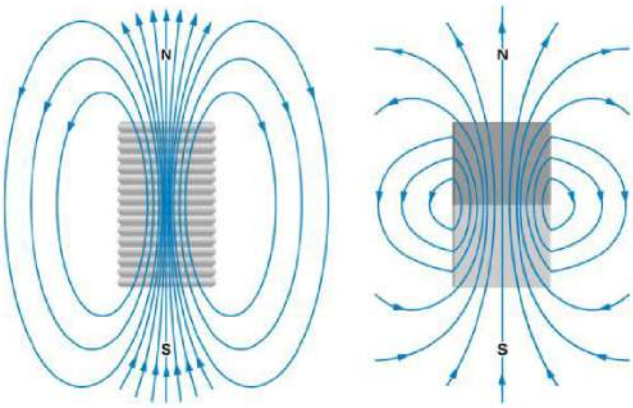
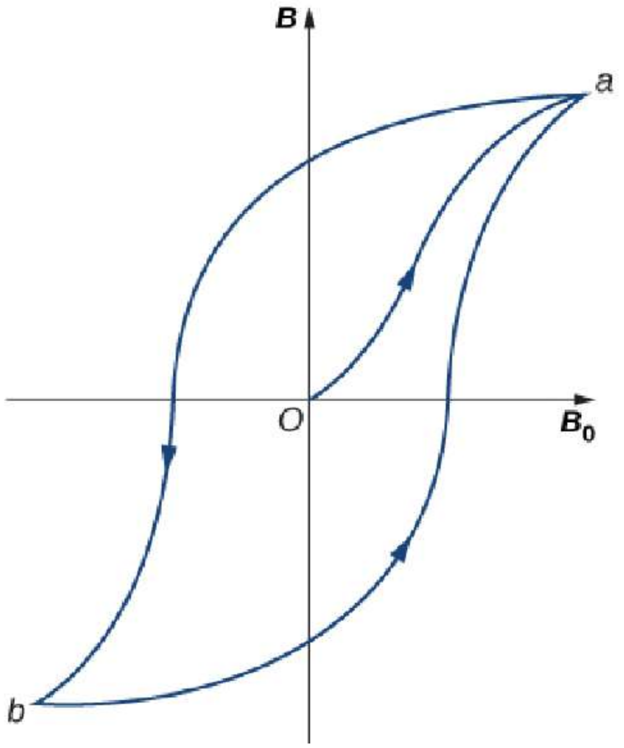


Figure 31: Comparison of the magnetic fields of a finite solenoid and a bar magnet



A typical hysteresis loop for a ferromagnet. When the material is first magnetized, it follows the curve from O to a. When B_0 is reversed, it takes the path shown from a to b. If B_0 is reversed again, the material follows the curve from b to a.

15.4 Applications: Hall Effect and Mass Spectrometry

The Hall Effect

A conductor carrying current through a magnetic field experiences a sideways magnetic force on its charge carriers. This produces a transverse electric field (Hall field) and a measurable voltage difference across the conductor.

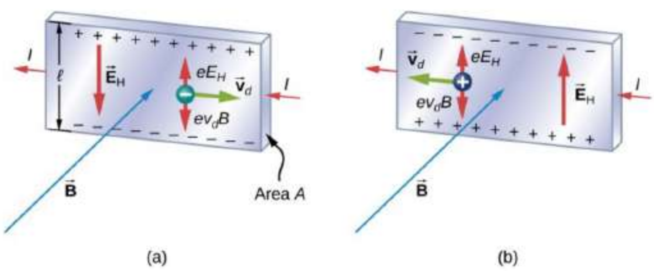


Figure 32: Hall voltage produced by magnetic deflection of charge carriers.

Drift velocity is found by balancing electric and magnetic forces:

$$eE_H = ev_dB \quad \Rightarrow \quad v_d = \frac{E_H}{B}$$

Since the Hall field is related to Hall voltage by $V_H = E_H\,\ell$, we obtain:

$$V_H = v_dB\,\ell$$

Using the current relation $I = nev_d A$, we eliminate v_d:

$$V_H = \frac{IB\ell}{neA}$$

Hall Voltage

$$V_H = \frac{IB\ell}{neA}$$

The Hall Probe

A Hall probe uses the Hall effect to measure magnetic field strength. The measured Hall voltage is proportional to the magnetic field:

$$V_H = \frac{IB\ell}{neA}$$

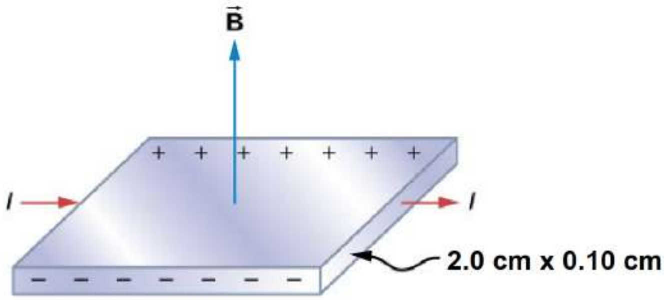


Figure 33: A Hall probe: current through a thin sample produces a Hall voltage proportional to B.

Velocity Selector and Mass Spectrometer

A velocity selector uses crossed electric and magnetic fields so that only ions with a specific velocity $v = E/B$ pass through without deflection.

After the selector, the ions enter a region with only a magnetic field B_0, undergoing circular motion.

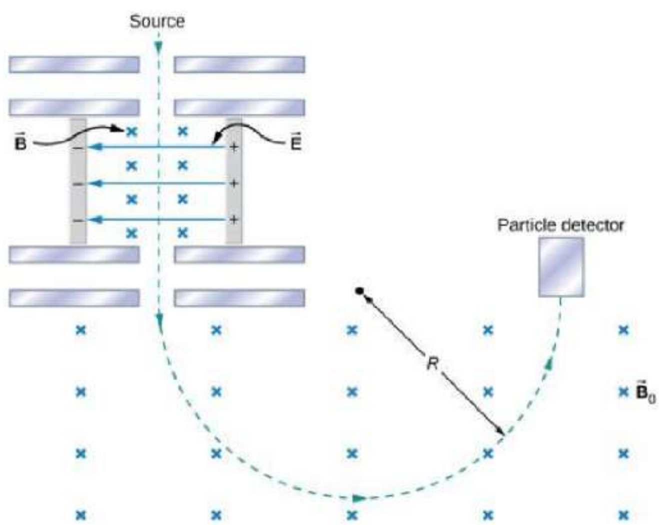


Figure 34: Bainbridge mass spectrometer: velocity selector + circular motion region.

Velocity selector condition:

$$v = \frac{E}{B}$$

Circular motion in magnetic field:

$$R = \frac{mv}{qB_0}$$

Combining:

$$R = \frac{m}{qB_0} \cdot \frac{E}{B}$$

Note! R is the radius not the diameter!!!

Charge-to-Mass Ratio (Bainbridge)

$$\frac{q}{m} = \frac{E}{BB_0R}$$

15.5 Magnetic Flux, Energy, and Gauss's Law

Magnetic Flux

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A} = B_\perp A$$

Units: $\mathrm{Wb} = \mathrm{T\,m^2}$.

Energy Stored in a Magnetic Field

For an inductor or solenoid:

$$U_B = \frac{1}{2}LI^2$$

and per unit volume:

$$u_B = \frac{B^2}{2\mu_0}$$

Gauss's Law for Magnetism

Magnetic monopoles do not exist; magnetic field lines always form closed loops.

$$\boxed{\oint \mathbf{B} \cdot d\mathbf{A} = 0}$$

15.6 Summary of Magnetic Relationships

$$
\begin{aligned}
v &= \frac{E}{B} \\
v_d &= \frac{E_H}{B} \\
V &= \frac{IB\ell}{neA} = B\ell v_d \\
\frac{q}{m} &= \frac{E}{BB_0R} \\
\mathbf{F} &= q(\mathbf{v} \times \mathbf{B}) \\
\mathbf{F} &= I\,\ell \times \mathbf{B} \\
\tau &= \mu \times \mathbf{B} \\
U &= -\mu \cdot \mathbf{B} \\
B &= \frac{\mu_0 I}{2\pi r}, \quad B_{\text{solenoid}} = \mu_0 n I, \quad B_{\text{toroid}} = \frac{\mu_0 N I}{2\pi r}
\end{aligned}
$$

16 Electromagnetic Induction

A changing magnetic flux through a conducting loop induces an emf. This phenomenon underlies the operation of generators, transformers, electric motors, and many electromagnetic devices.

16.1 Magnetic Flux

Magnetic flux measures how much magnetic field penetrates a surface:

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A} = B_\perp A$$

where $B_\perp$ is the component of $\mathbf{B}$ perpendicular to the surface.

Units: Wb ($\mathrm{T\,m^2}$).

16.2 Faraday's Law of Induction

A time-varying magnetic flux induces an emf in a closed loop.

Faraday's Law

$$\varepsilon = -\frac{d\Phi_B}{dt} \qquad \text{(single loop)}$$

$$\varepsilon = -N\,\frac{d\Phi_B}{dt} \qquad \text{(coil with } N \text{ turns)}$$

The induced emf increases when:

- the magnetic field changes faster,
- the area changes,
- or the orientation of the loop changes.

16.3 Lenz's Law (Direction of Induced Current)

Faraday's law gives the magnitude of the induced emf. The minus sign incorporates Lenz's law:

> *The induced current flows in a direction that produces a magnetic field opposing the change in flux.*

This ensures conservation of energy.

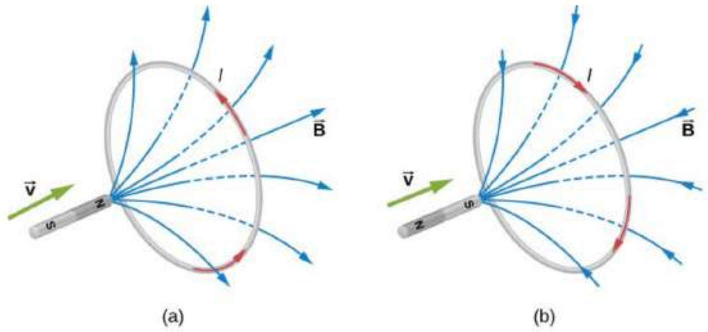


Figure 35: Lenz's law: the induced current opposes the change in magnetic flux.

Sign convention (Lenz's law): A positive induced emf corresponds to a current whose magnetic field *opposes the increase in flux.* If the flux decreases, the induced current produces a field that *reinforces the original direction.*

16.4 Typical Induction Situations

Common scenarios involving changing flux:

- Moving a magnet toward or away from a coil.
- Moving the coil instead of the magnet.
- Changing the magnetic field strength in time.
- Rotating the loop in a magnetic field.

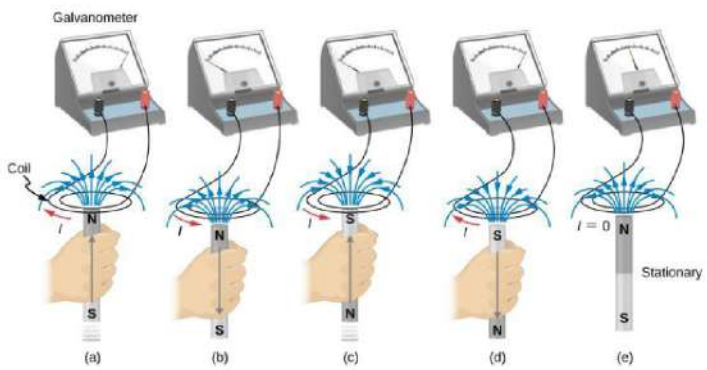


Figure 36: Examples of induction scenarios involving relative motion or changing magnetic fields.

16.5 Motional emf

A conducting rod of length ℓ moving with velocity v through a magnetic field B develops an emf due to the magnetic force on charge carriers:

$$\varepsilon = B\ell v$$

This result is consistent with Faraday's law applied to a loop whose area changes in time.

Motional emf

$$\varepsilon = B\ell v$$

16.6 emf of a Rotating Coil (Electric Generator)

A generator consists of a coil of area A with N turns rotating in a uniform magnetic field B with angular velocity ω.

The magnetic flux through the coil is

$$\Phi_B(t) = NBA\cos(\omega t)$$

Faraday's law gives the induced emf:

$$\varepsilon(t) = -\frac{d\Phi_B}{dt} = NBA\omega\sin(\omega t)$$

Generator emf

$$\varepsilon_{\max} = NBA\omega \qquad \varepsilon(t) = \varepsilon_{\max}\sin(\omega t)$$

This is the fundamental equation of AC generator output.

Power and Torque in a Generator

The mechanical power required to rotate a coil that produces an emf $\varepsilon(t)$ and current I in the external circuit is

$$P_{\text{mech}} = I\,\varepsilon.$$

Since mechanical power is also $P_{\text{mech}} = \tau\omega$, the required torque is

$$\tau = \frac{I\,\varepsilon}{\omega}.$$

16.7 Energy and Work in Induction

Induced currents oppose the change in flux (Lenz's law). Therefore external work must be done to keep changing the flux:

- pushing a magnet into a coil requires force,
- pulling it out requires force in the opposite direction,
- rotating a coil in a generator requires torque.

Mechanical work done becomes electrical energy.

16.8 Motional emf in Practical Systems

A conducting rod of length ℓ moving with velocity v through a magnetic field B experiences an induced emf:

$$\varepsilon = B\ell v$$

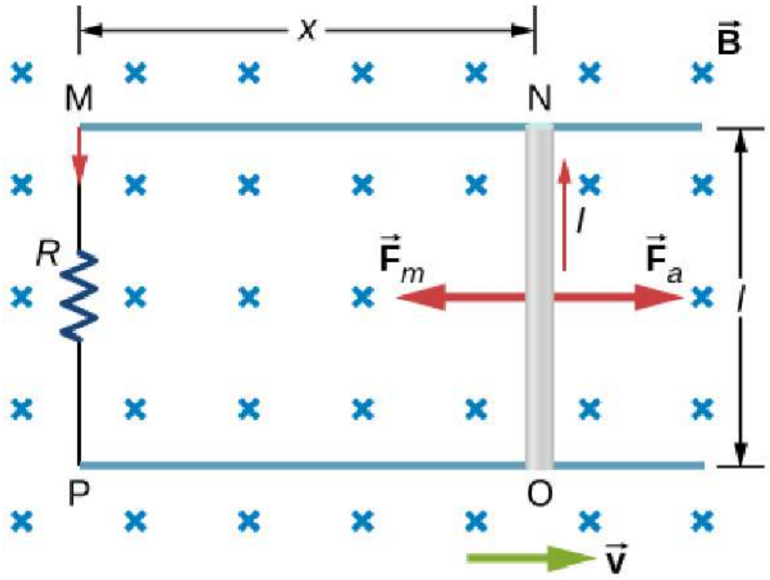


Figure 37: A moving conducting rod in a magnetic field produces a motional emf.

Faraday's law applied to a moving-loop geometry leads to the same result.

Motional emf (again)

$$\varepsilon = B\ell v$$

Example: The Tethered Satellite Generator

A long conducting tether dragged through Earth's magnetic field at orbital speed generates a significant emf, converting mechanical to electrical energy.

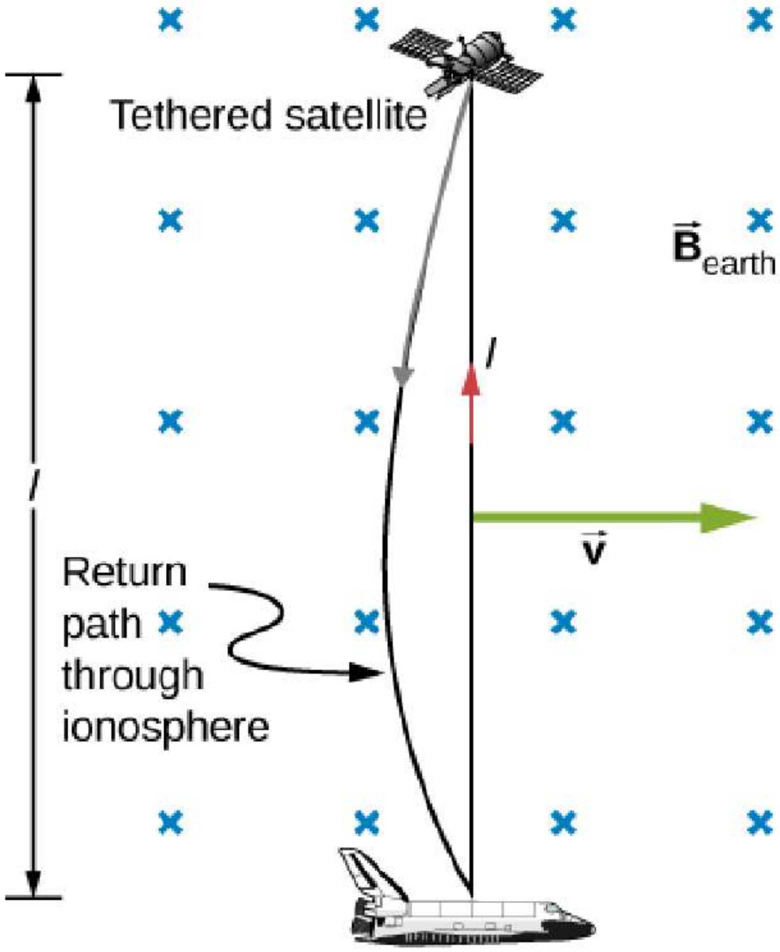


Figure 38: Tethered satellite experiment using motional emf for power generation.

16.9 Induced Electric Fields

A changing magnetic field induces an electric field even in empty space:

Induced Electric Field

$$\oint \mathbf{E} \cdot d\ell = -\frac{d\Phi_B}{dt}$$

Example: a decreasing current in a solenoid induces circular electric fields both inside and outside the solenoid.

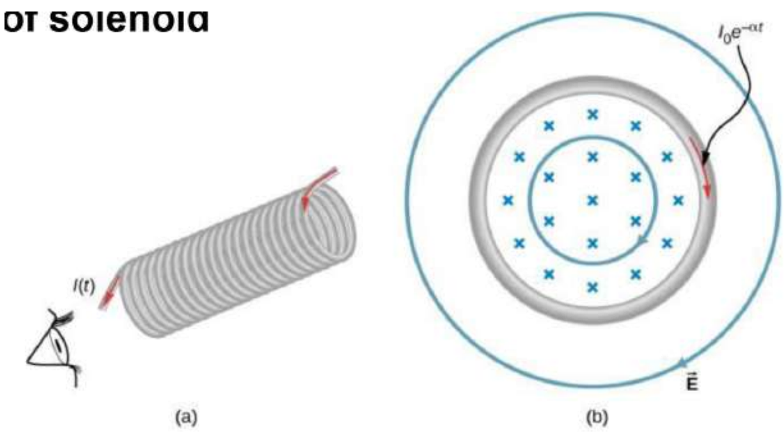


Figure 39: Changing magnetic field in a solenoid produces induced electric field loops.

Summary of Induction Formulas

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}, \qquad \varepsilon = -\frac{d\Phi_B}{dt}, \qquad \varepsilon_{\text{motion}} = B\ell v,$$

$$\varepsilon_{\text{gen}}(t) = NBA\omega \sin(\omega t), \qquad \oint \mathbf{E} \cdot d\ell = -\frac{d\Phi_B}{dt}.$$

17 Appendix: Constants and Quick Reference

Fundamental Constants

Quantity	Symbol	Value (SI)
Speed of light	c	2.998×10^8 m/s
Gravitational constant	G	6.674×10^{-11} N m^2/kg^2
Permittivity of free space	ε_0	8.854×10^{-12} C^2/(N m^2)
Permeability of free space	μ_0	$4\pi \times 10^{-7}$ N/A^2
Elementary charge	e	1.602×10^{-19} C
Electron mass	m_e	9.109×10^{-31} kg
Proton mass	m_p	1.673×10^{-27} kg
Avogadro's number	N_A	6.022×10^{23} 1/mol
Boltzmann constant	k_B	1.381×10^{-23} J/K
Universal gas constant	R	8.314 J/(mol K)
Planck constant	h	6.626×10^{-34} J s
Stefan–Boltzmann constant	σ	5.670×10^{-8} W/(m^2 K^4)

Unit Conversions and Prefixes

Mechanical:

- 1 N = 1 kg m/s^2
- 1 J = 1 N m
- 1 W = 1 J/s
- 1 Pa = 1 N/m^2

Thermal:

- 1 cal = 4.186 J
- 1 atm = 1.013×10^5 Pa
- 0 °C = 273.15 K

Electrical:

- 1 C = 1 A s
- 1 V = 1 J/C

- $1\ \Omega = 1$ V/A
- 1 F $= 1$ C/V
- 1 T $= 1$ N/(A m)

Volume:

- $1\ m^3 = 1000\ L$
- $1\ dm^3 = 1\ L$
- $1\ cm^3 = 1\ ml$

Common Prefixes:

Prefix	Symbol	Factor
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}

Thermodynamic Laws Summary

Law	Statement
0th Law	Defines temperature through thermal equilibrium.
1st Law	$\Delta U = Q - W$ (Energy conservation).
2nd Law	Entropy of an isolated system never decreases; $\Delta S_{\text{universe}} \geq 0$.
3rd Law	As $T \to 0$ K, $S \to 0$.

Quick Formulas by Topic

Kinematics
$x = x_0 + v_0 t + \frac{1}{2}at^2$ $v = v_0 + at$ $v^2 = v_0^2 + 2a\Delta x$

Forces
$\sum \mathbf{F} = m\mathbf{a}$ $f_k = \mu_k N,\ f_s \leq \mu_s N$ $F_{\text{spring}} = -k\Delta x$

Energy
$K = \frac{1}{2}mv^2,\ U_g = mgh$ $P = \mathbf{F}\cdot\mathbf{v}$ $E_{\text{mech}} = K + U$

Momentum
$\mathbf{p} = m\mathbf{v},\ \mathbf{J} = \Delta\mathbf{p}$ $\sum \mathbf{p}_i = \sum \mathbf{p}_f$

Rotation
$\tau = I\alpha,\ L = I\omega$ $K_{\text{rot}} = \frac{1}{2}I\omega^2$ $I_{\text{rod,center}} = \frac{1}{12}ML^2$

Fluids
$p = p_0 + \rho g h$ $F_B = \rho V g$ $p + \frac{1}{2}\rho v^2 + \rho g y = \text{const}$

Electrostatics
$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$ $E = \frac{F}{q}$ $V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$

Circuits
$V = IR,\quad P = IV = I^2R = \frac{V^2}{R}$ $R_{\text{series}} = \sum R_i,\ 1/R_{\text{parallel}} = \sum 1/R_i$

Magnetism
$\mathbf{F} = q\mathbf{v}\times\mathbf{B}$ $\mathbf{F} = I\boldsymbol{\ell}\times\mathbf{B}$ $\tau = \mu\times\mathbf{B}$

Gravitation
$F = G\frac{m_1 m_2}{r^2}$ $U = -G\frac{Mm}{r}$ $v_{\text{esc}} = \sqrt{2GM/R}$

Thermodynamics
$\Delta U = Q - W$ $e_{\text{Carnot}} = 1 - \frac{T_c}{T_h}$ $S = k_B \ln\Omega$

Mathematical and Vector Identities

$$\mathbf{A}\cdot\mathbf{B} = AB\cos\theta$$
$$\mathbf{A}\times\mathbf{B} = AB\sin\theta\,\hat{\mathbf{n}}$$
$$\text{Work: } W = \mathbf{F}\cdot\Delta\mathbf{r}$$
$$\text{Torque: } \tau = \mathbf{r}\times\mathbf{F}$$

$$\sin^2\theta + \cos^2\theta = 1$$
$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$
$$1 + \tan^2\theta = \sec^2\theta$$

17.1 Trigonometric formulas

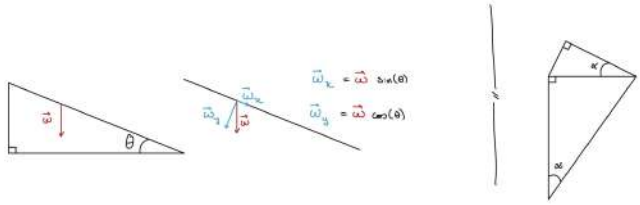


Figure 40: Trigonometric drawings for angled planes

$$\sin(\theta) = \frac{o}{h}, \qquad \cos(\theta) = \frac{a}{h}, \qquad \tan(\theta) = \frac{o}{a}$$

Where a is the adjacent cathetus, o is the opposite cathetus and h is the hypotenuse of the right triangle

17.2 Basic geometry formulas (used throughout physics)

Circle

$$A = \pi r^2, \qquad C = 2\pi r$$

Sphere

$$A = 4\pi r^2, \qquad V = \frac{4}{3}\pi r^3$$

Cylinder

$$A_{\text{cross}} = \pi r^2, \qquad A_{\text{side}} = 2\pi r h$$
$$A_{\text{total}} = 2\pi r^2 + 2\pi r h, \qquad V = \pi r^2 h$$

Use in physics

- Gauss' law: choose surface area based on symmetry.
- Density definitions:
$$\rho = \frac{m}{V}, \qquad \sigma = \frac{Q}{A}, \qquad \lambda = \frac{Q}{L}.$$
Here ρ is volume charge density (C/m^3), σ surface charge density (C/m^2), and λ linear charge density (C/m).
- Fluid flow: $Q = Av$.

Exam note Always identify the geometry and symmetry before inserting an area or volume into a physical law. A wrong geometric factor leads to a wrong result even if the physics is correct.

Compiled with LaTeX by Afonso Rodrigues de Brito and Marta Delgado Ordiales — November 2025